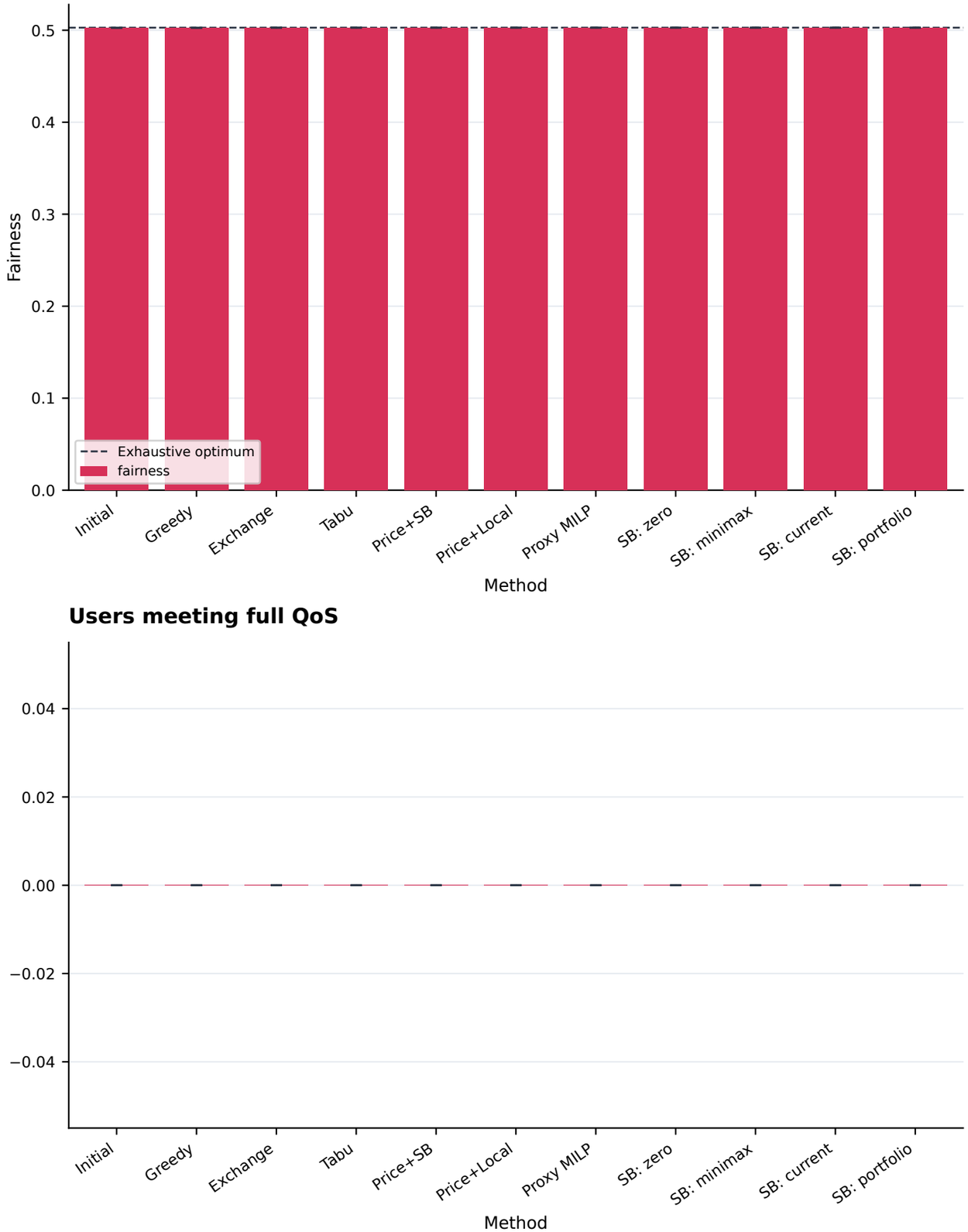
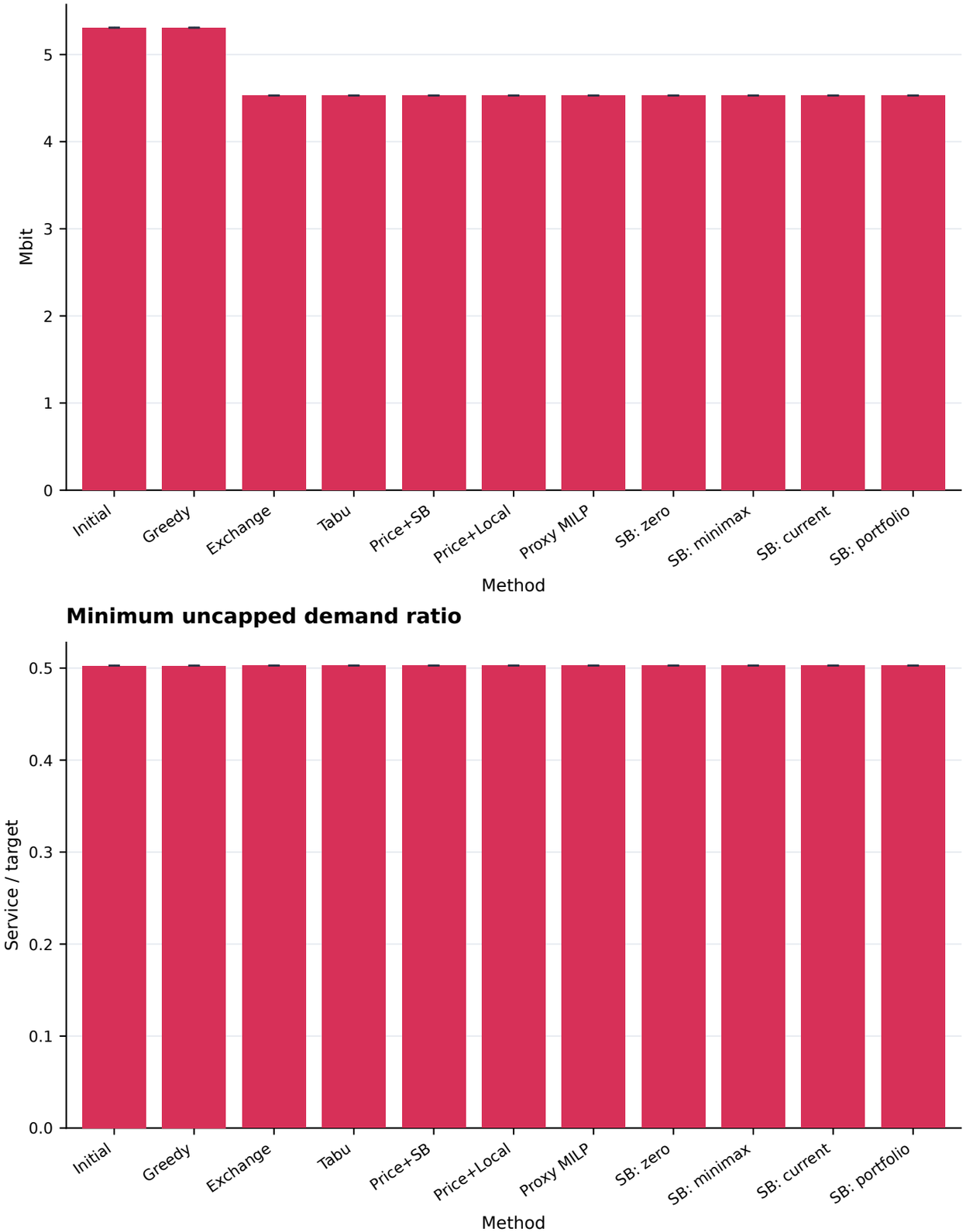
small | K=8, M=3, L=8 | median and min-max across 3 seeds

Capped minimum QoS ratio



Original net service

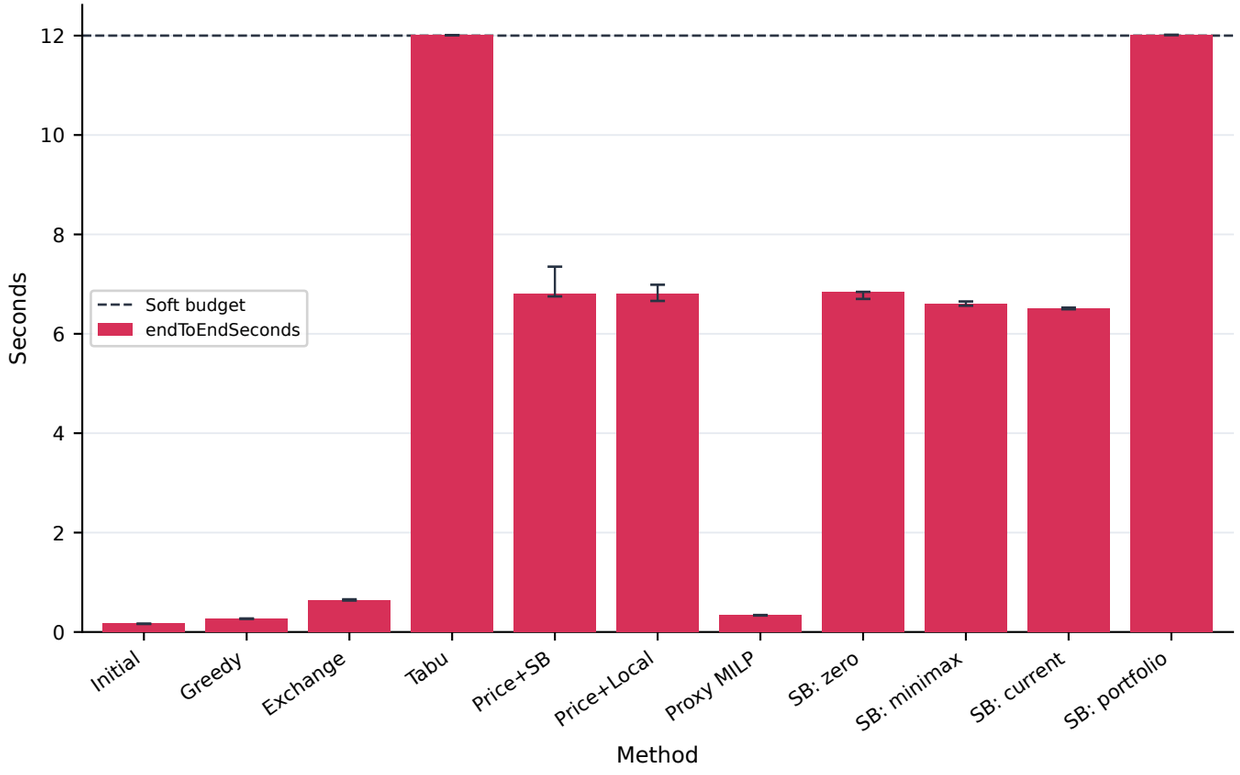


Users meeting full QoS

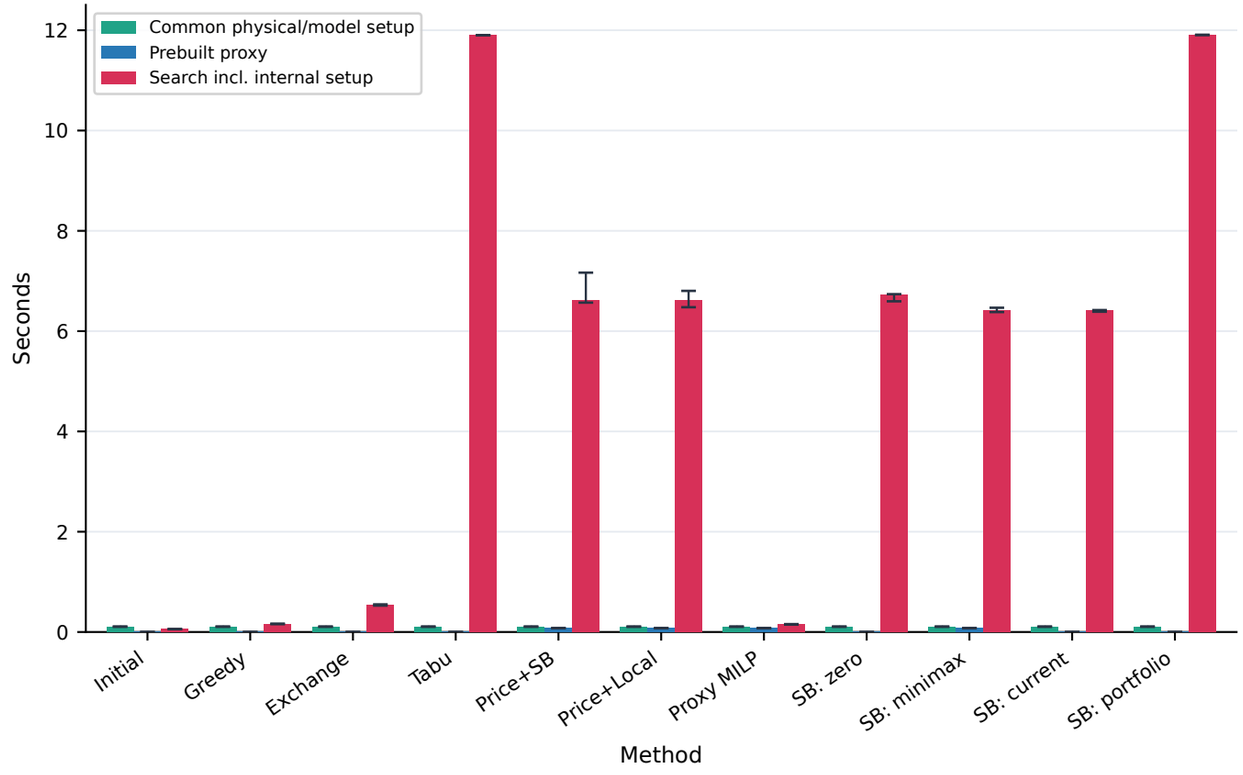
Minimum uncapped demand ratio

small | K=8, M=3, L=8 | median and min-max across 3 seeds

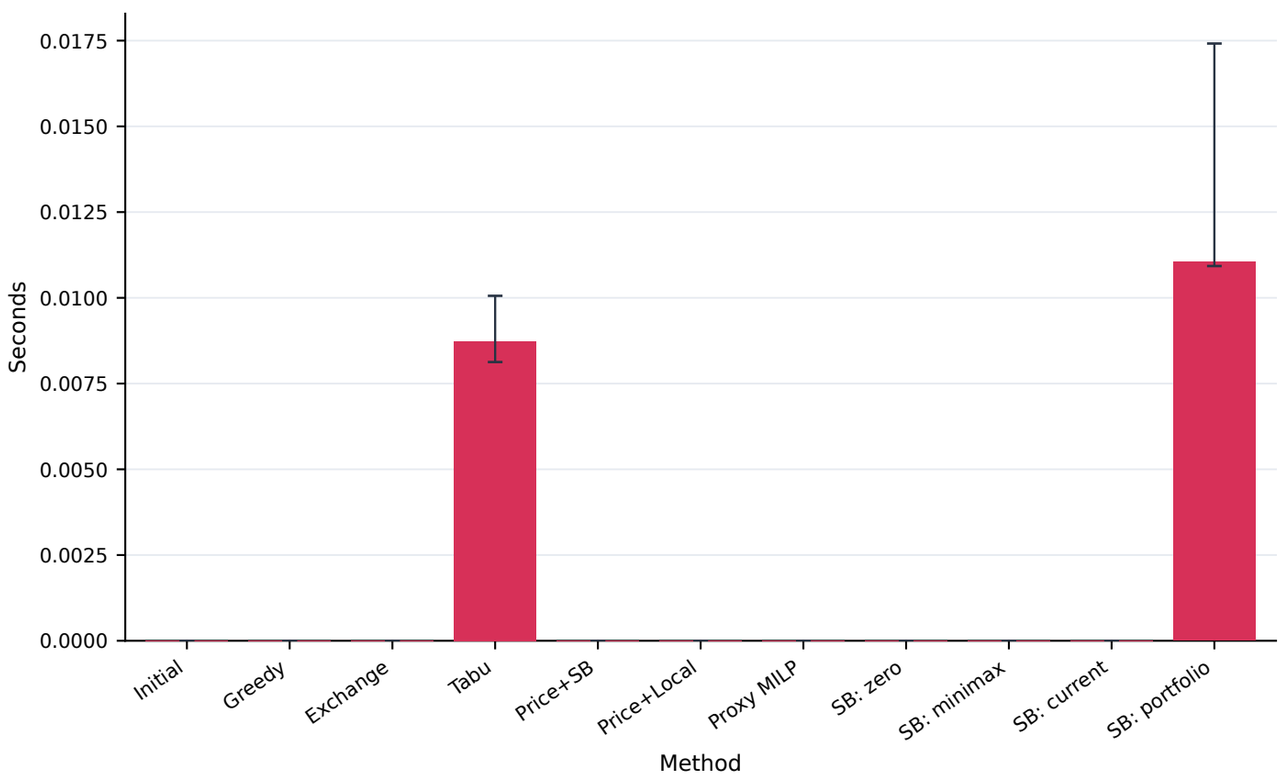
Accounted end-to-end time



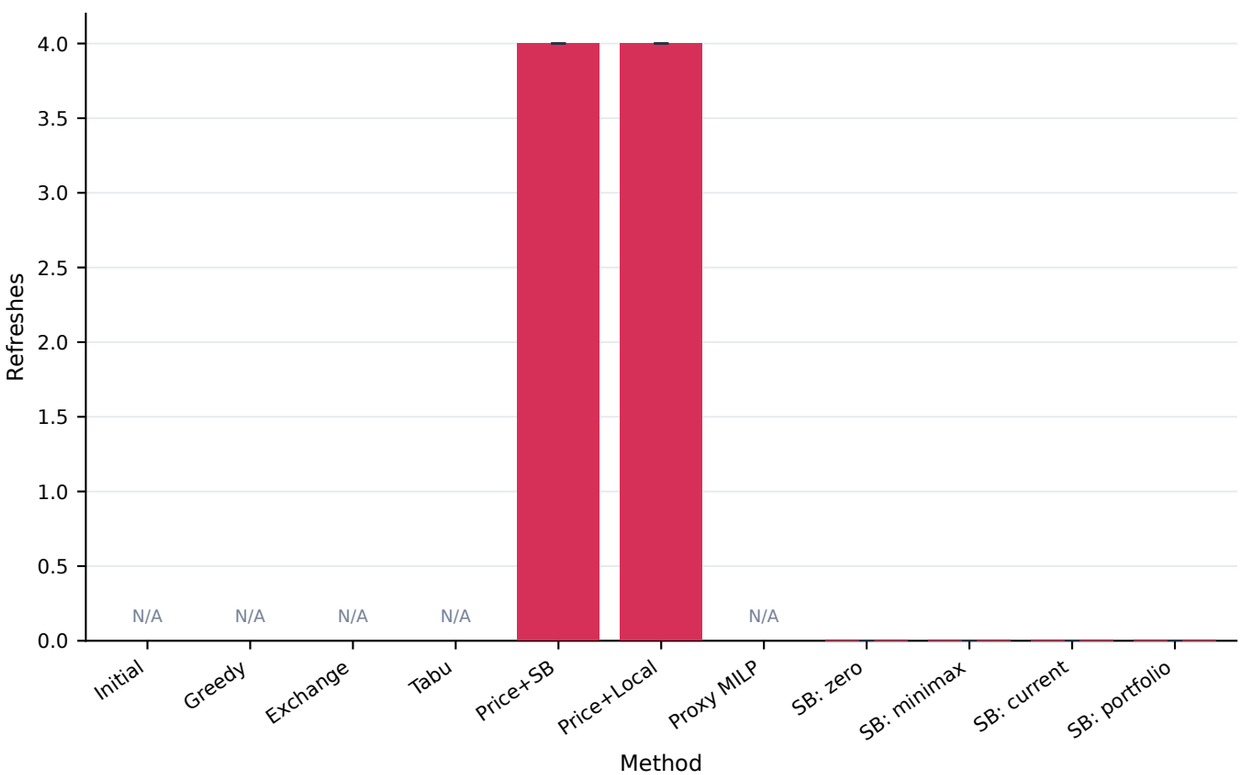
Timing components (internal proxy work stays in search)



Actual budget overrun (not clipped)

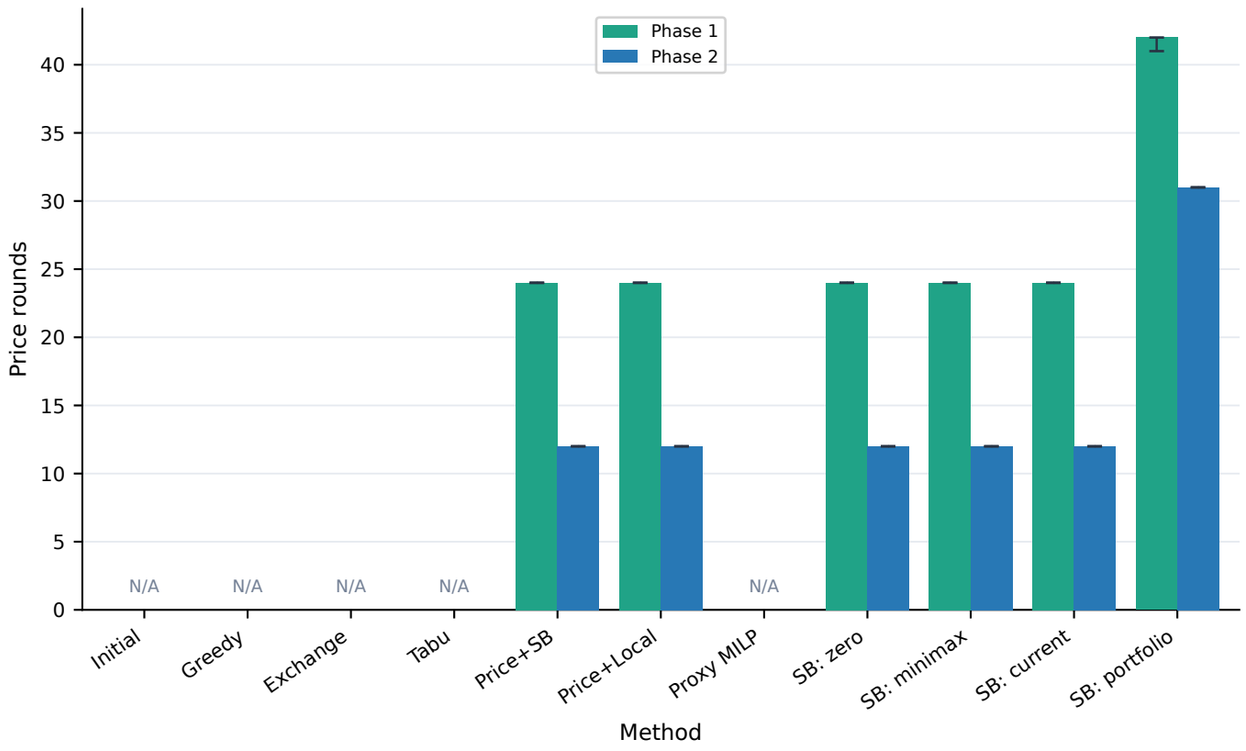


Proxy refresh count

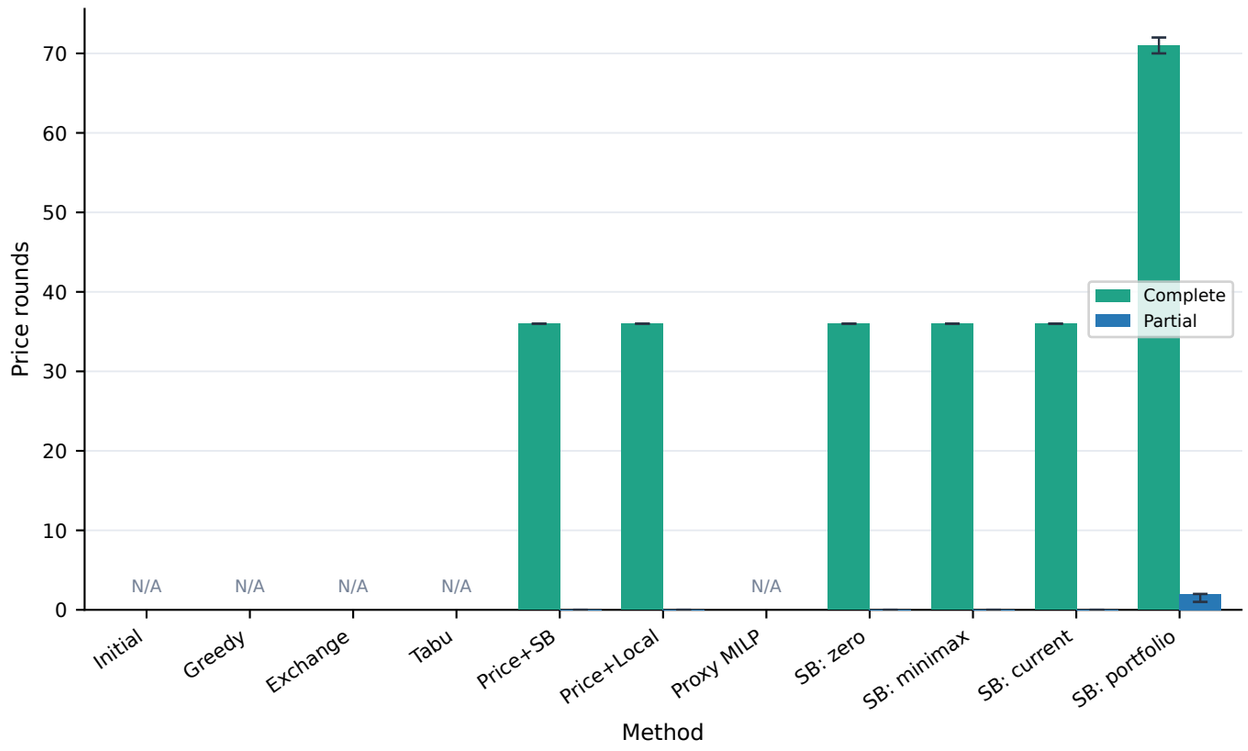


small | K=8, M=3, L=8 | median and min-max across 3 seeds

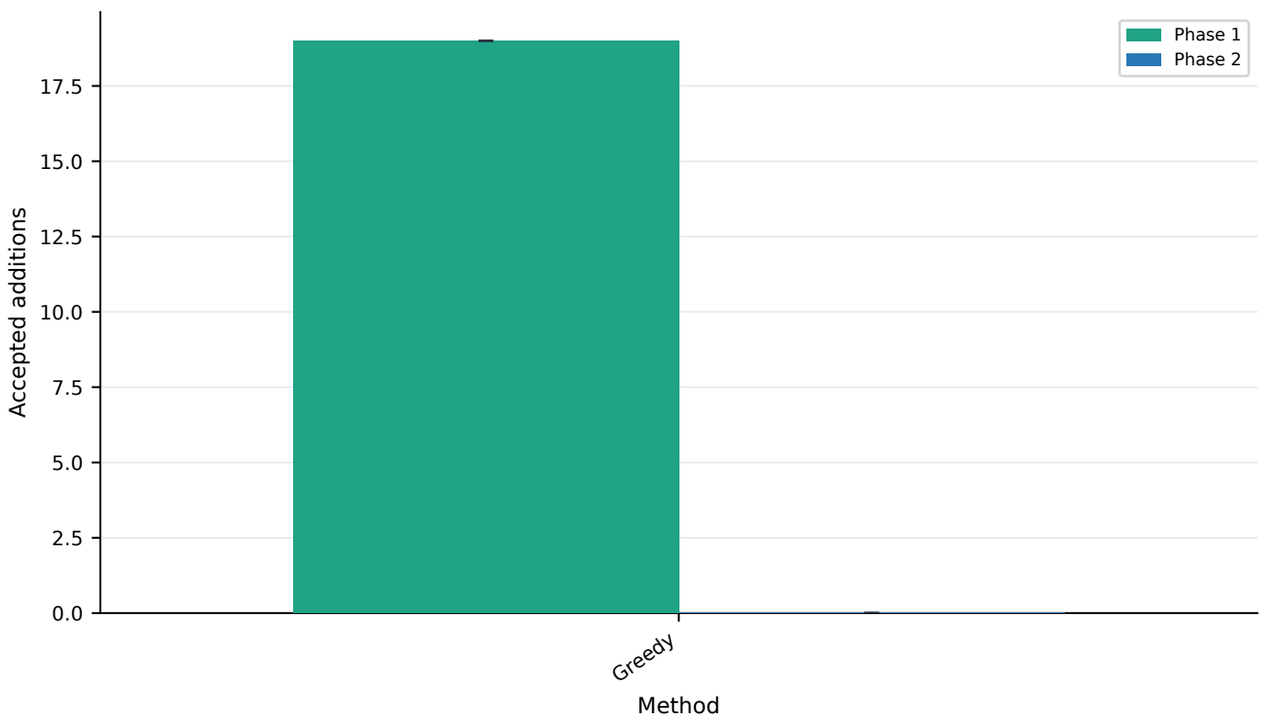
Price rounds attempted, including partial



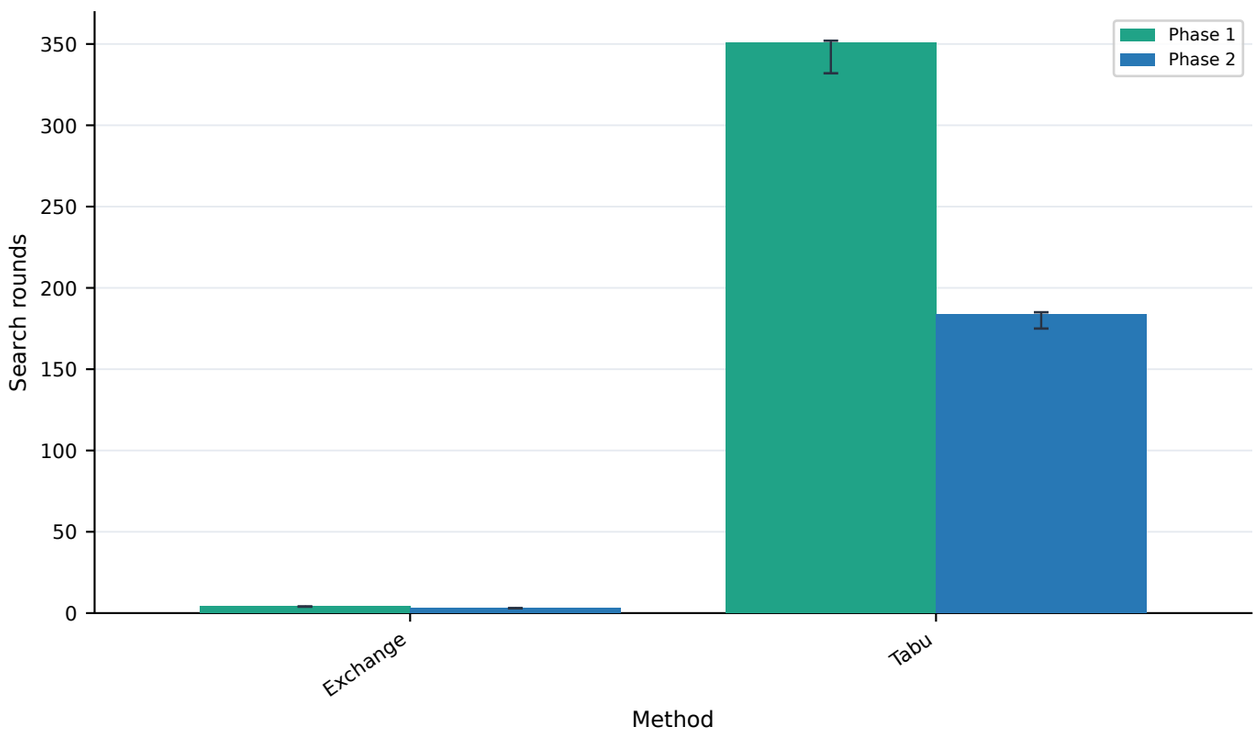
Complete versus partial price rounds



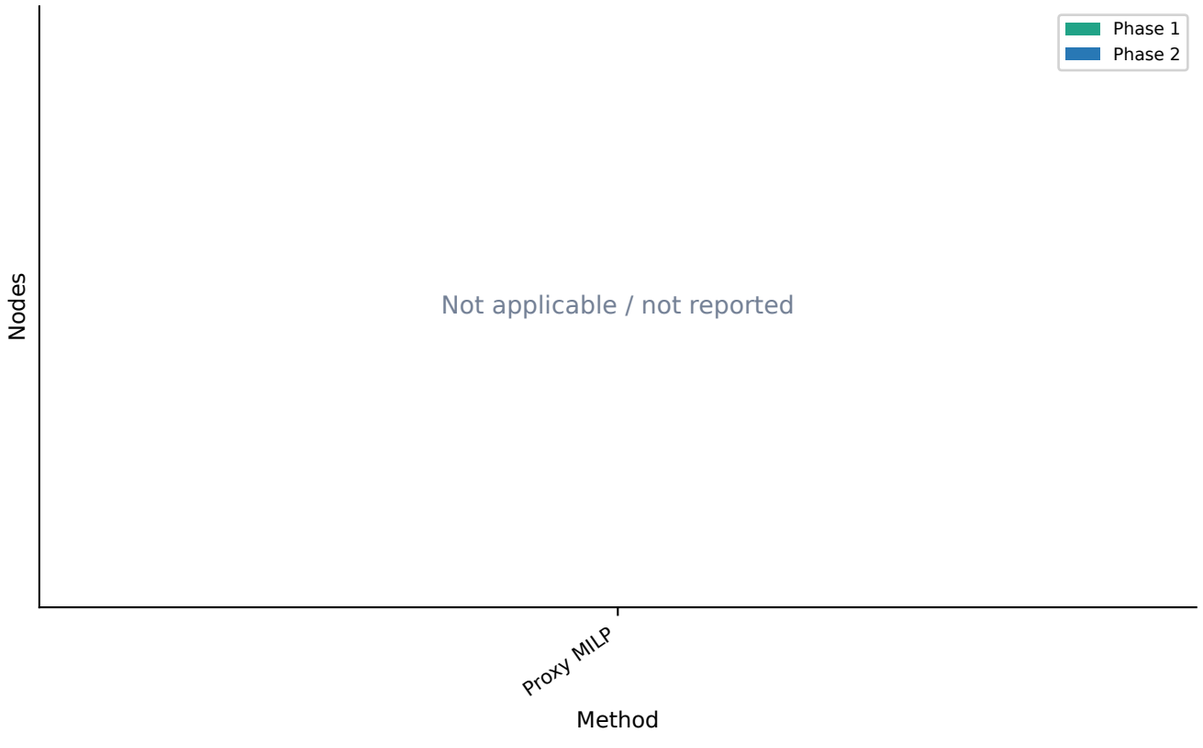
Greedy accepted additions



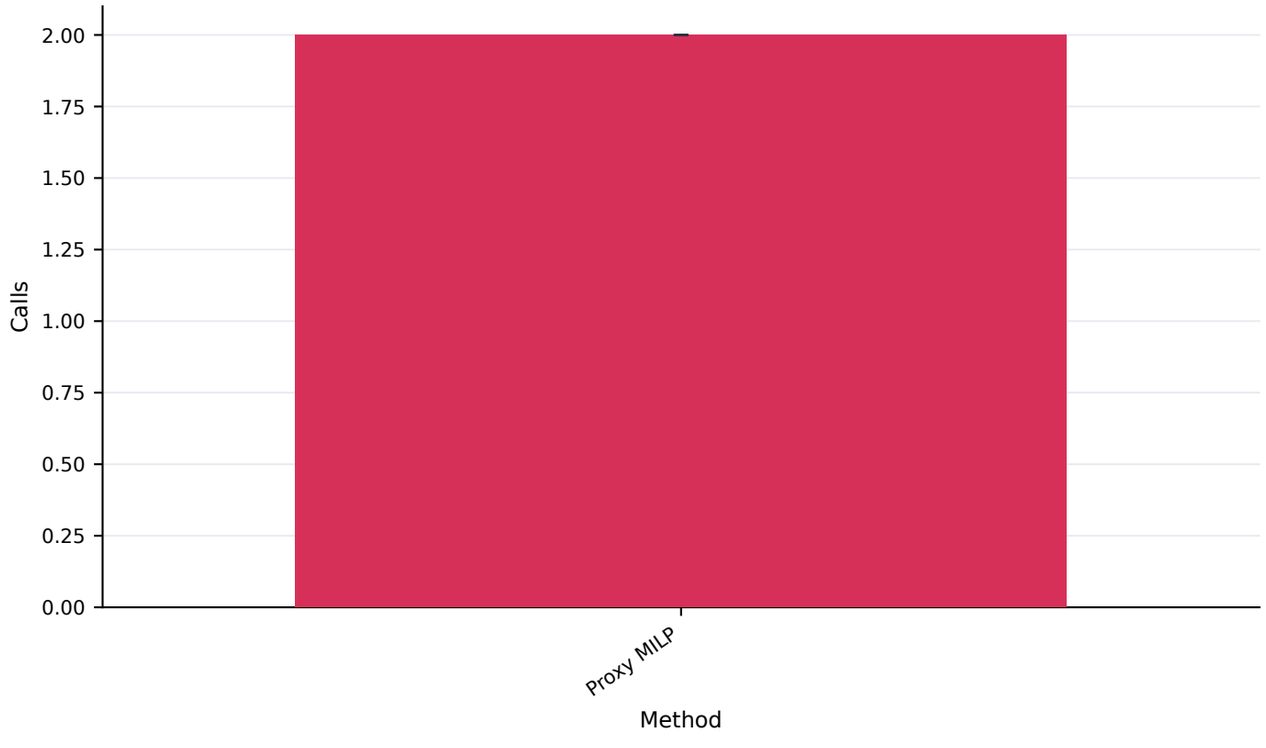
Original-objective neighborhood rounds



MILP branch-and-bound nodes



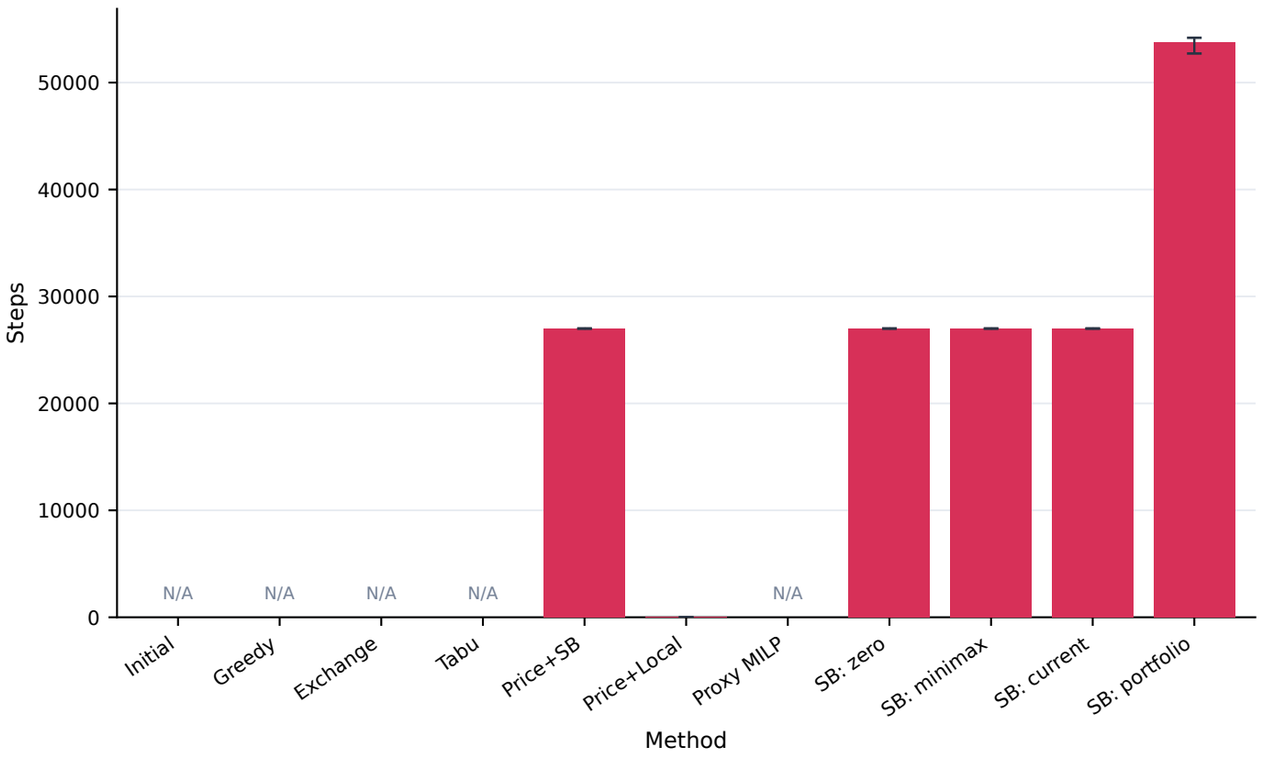
MILP solver invocations (separate unit)



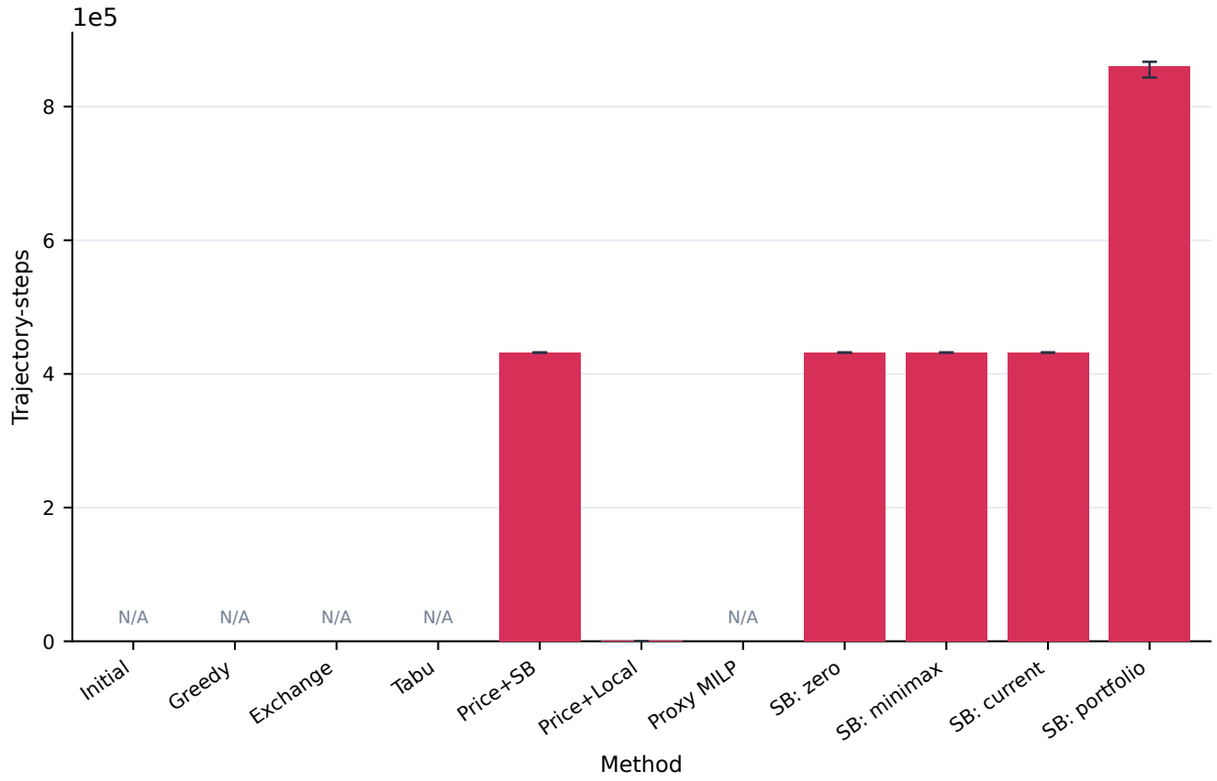
GLPK does not report node counts; missing values are not zero.

small | K=8, M=3, L=8 | median and min-max across 3 seeds

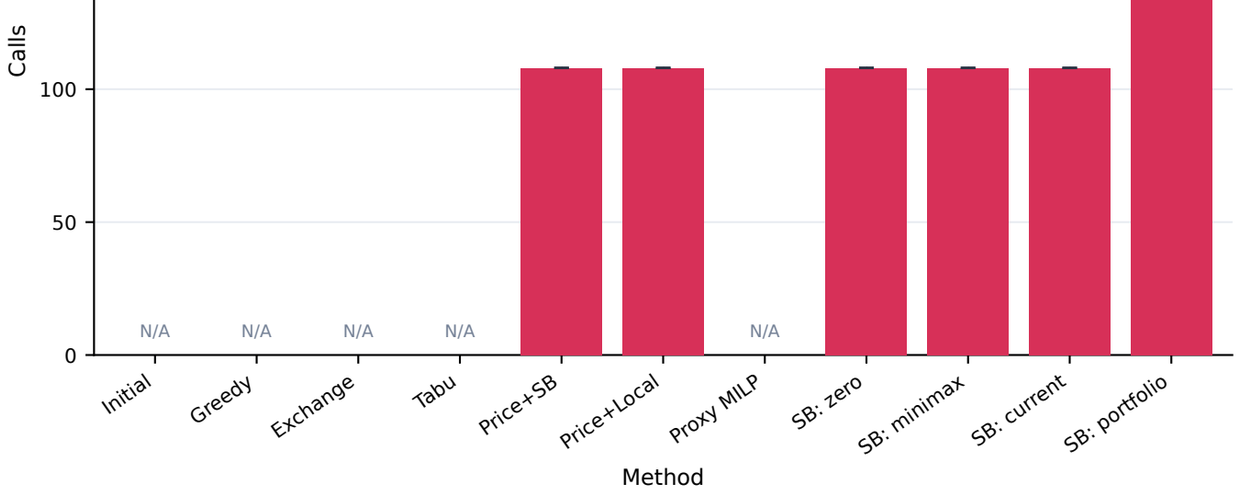
SB integration steps summed over all calls



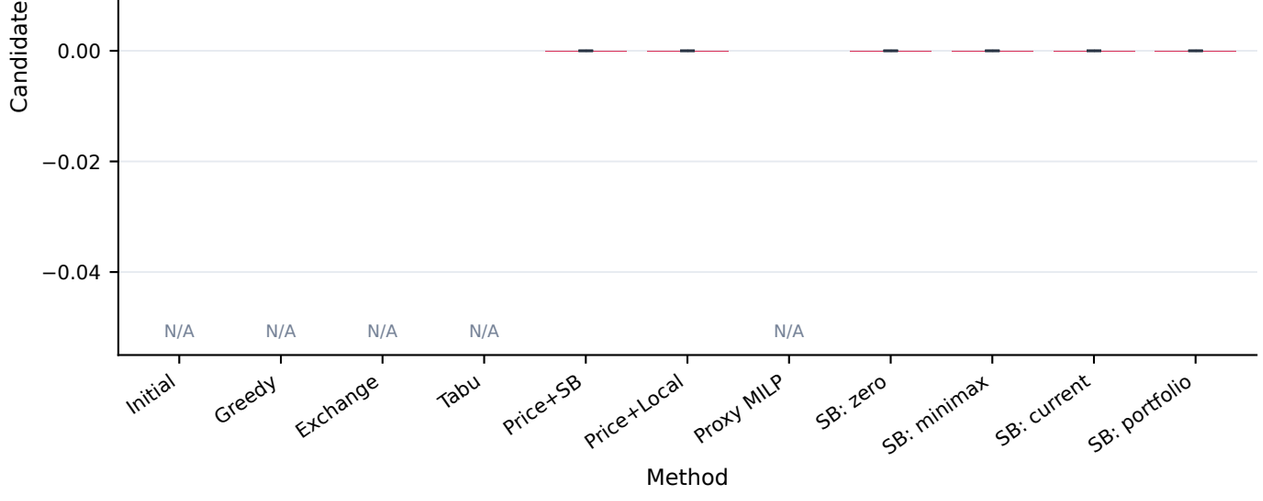
SB steps multiplied by trajectory count



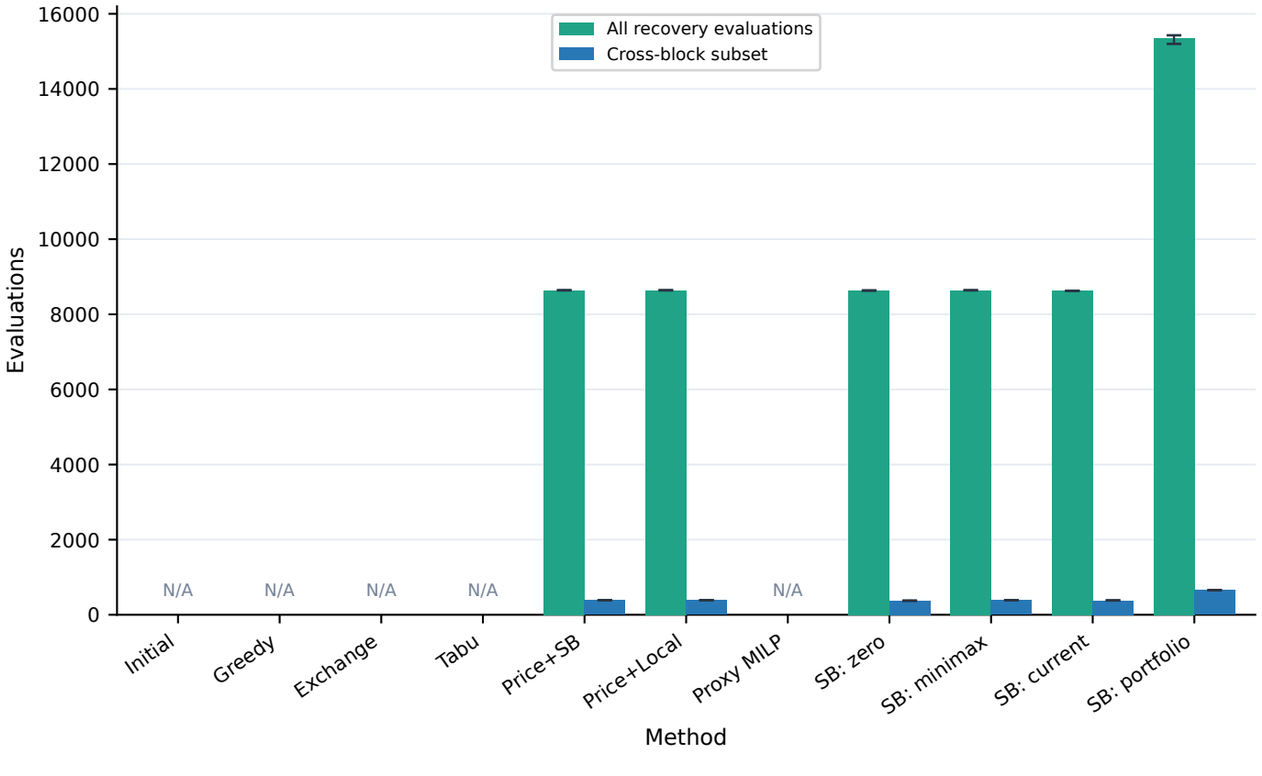
Resource subproblems solved



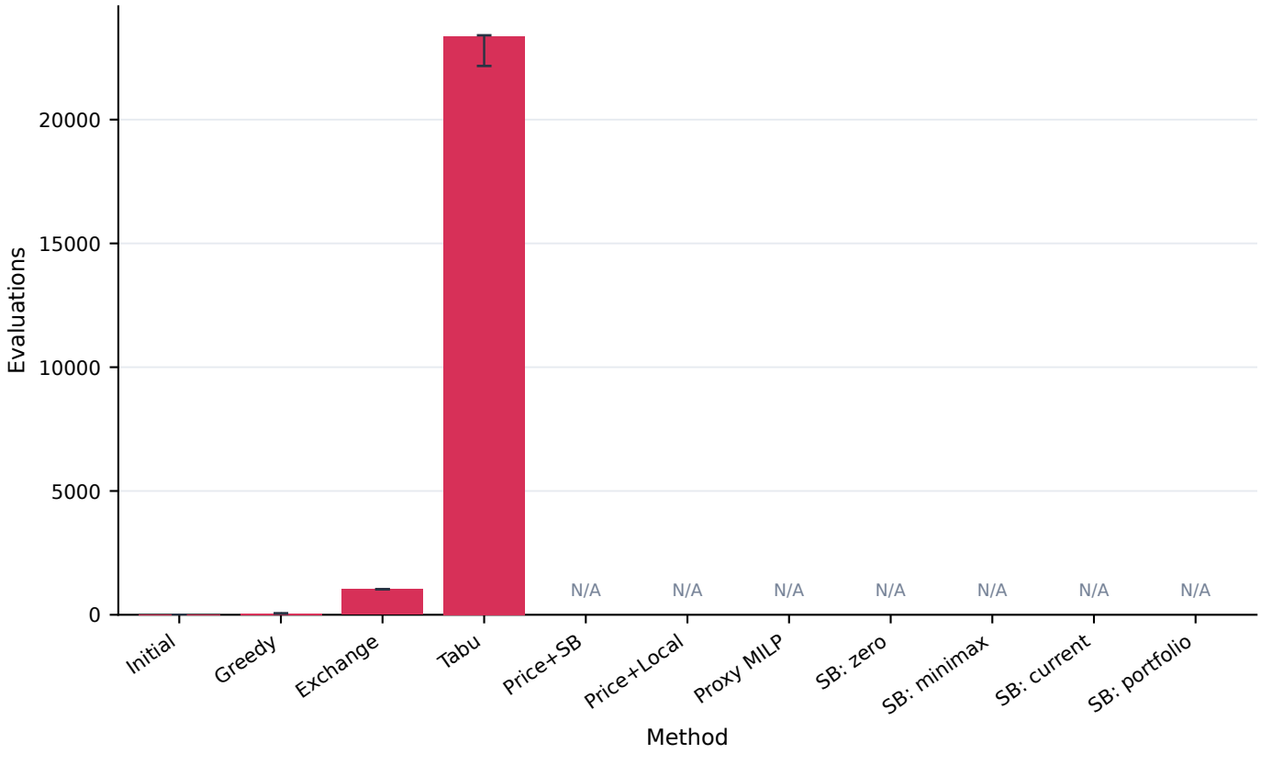
Candidates changed by hard repair



Recovery evaluations (cross is a subset)

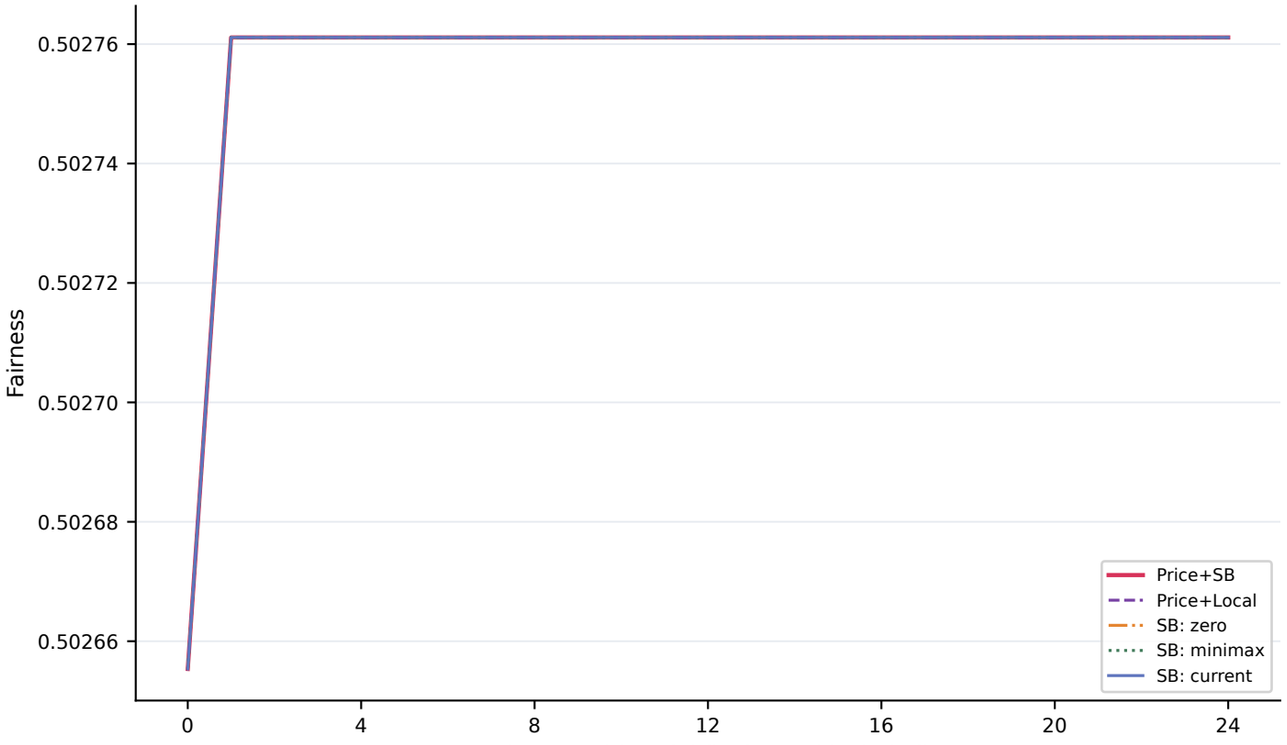


Original baseline search evaluations

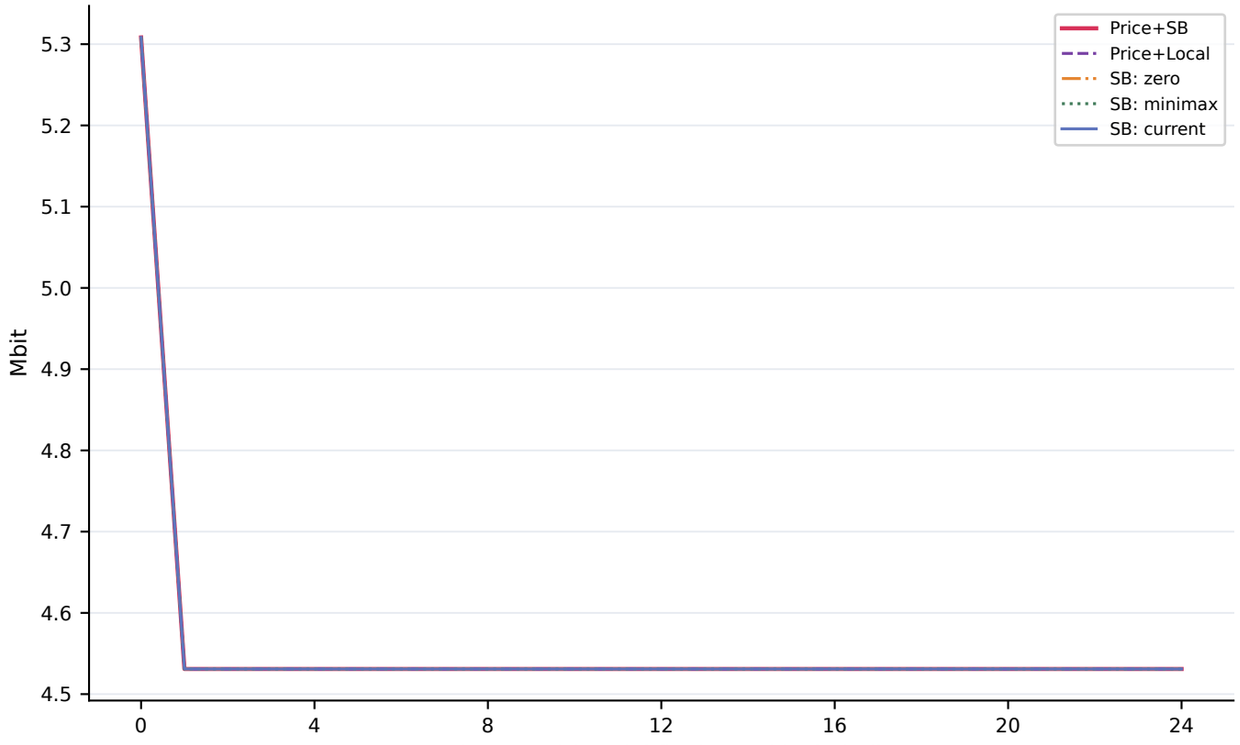


small | first requested seed 20261105 | price methods; portfolio branches excluded

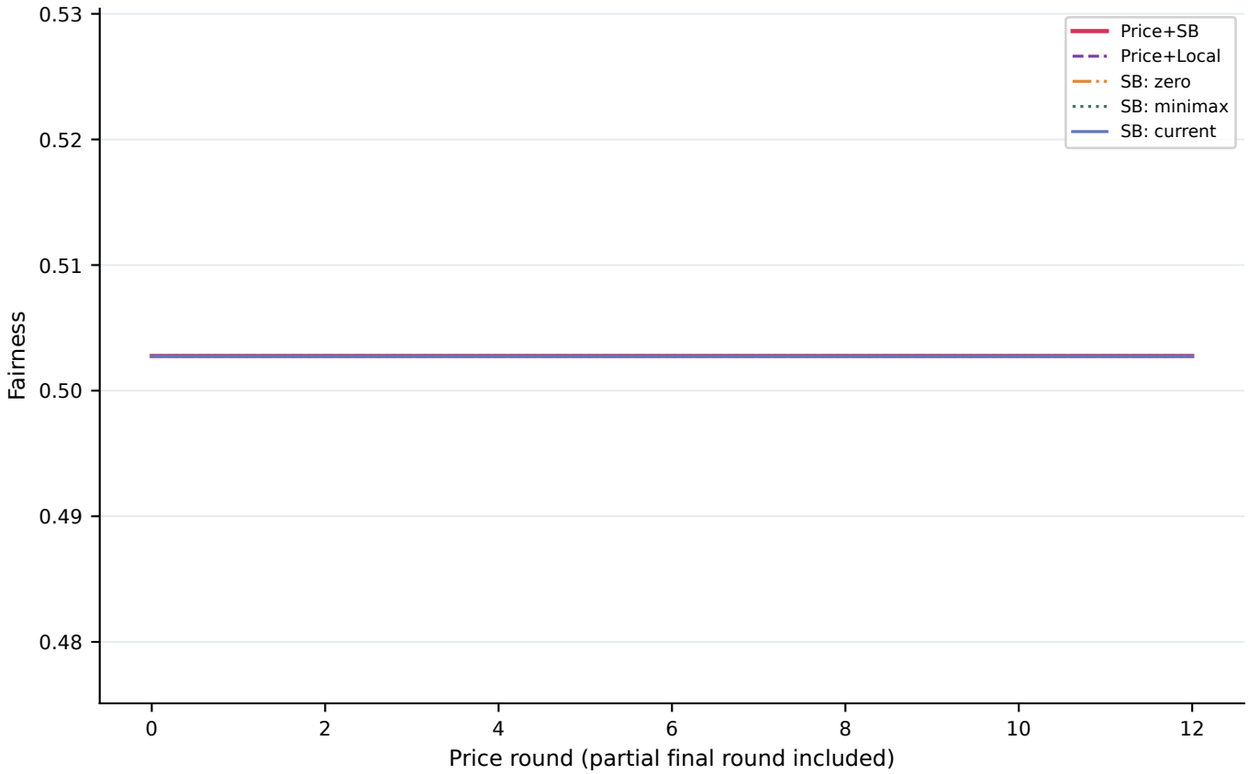
Saved original fairness: phase 1



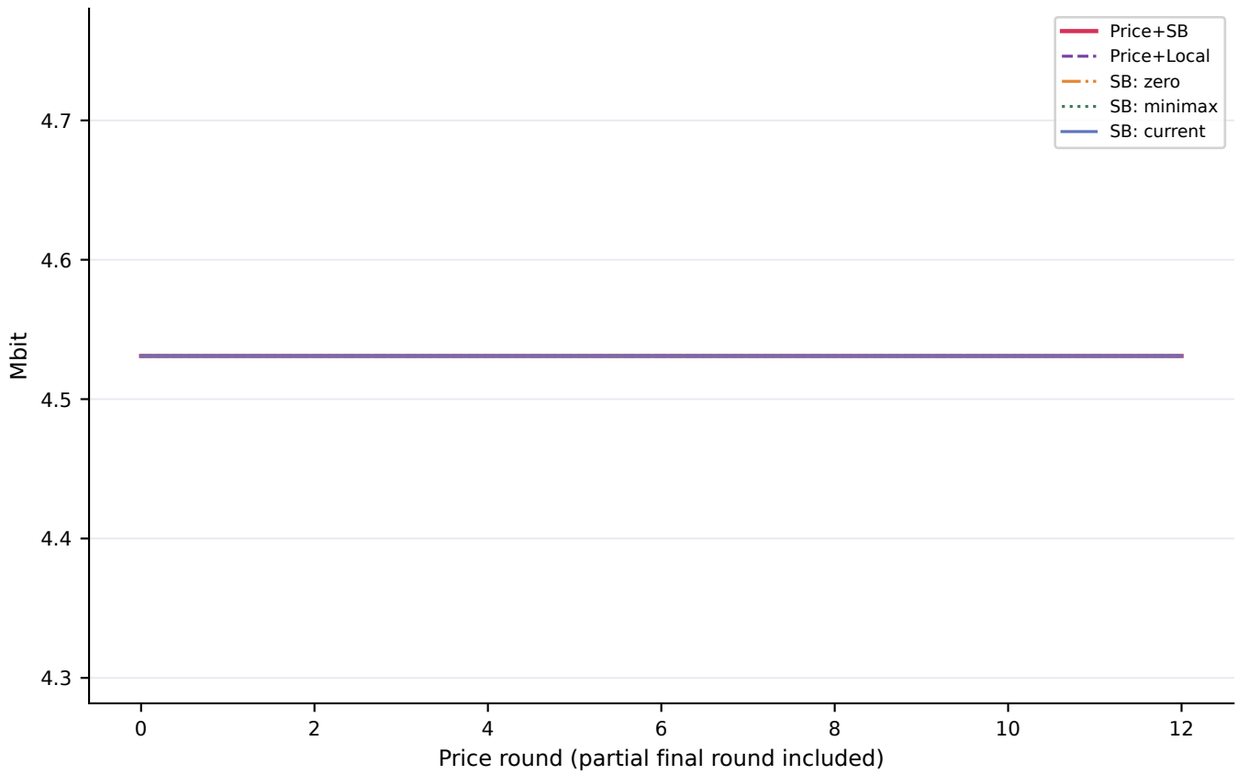
Saved throughput: phase 1 (may decrease)



Saved original fairness: phase 2

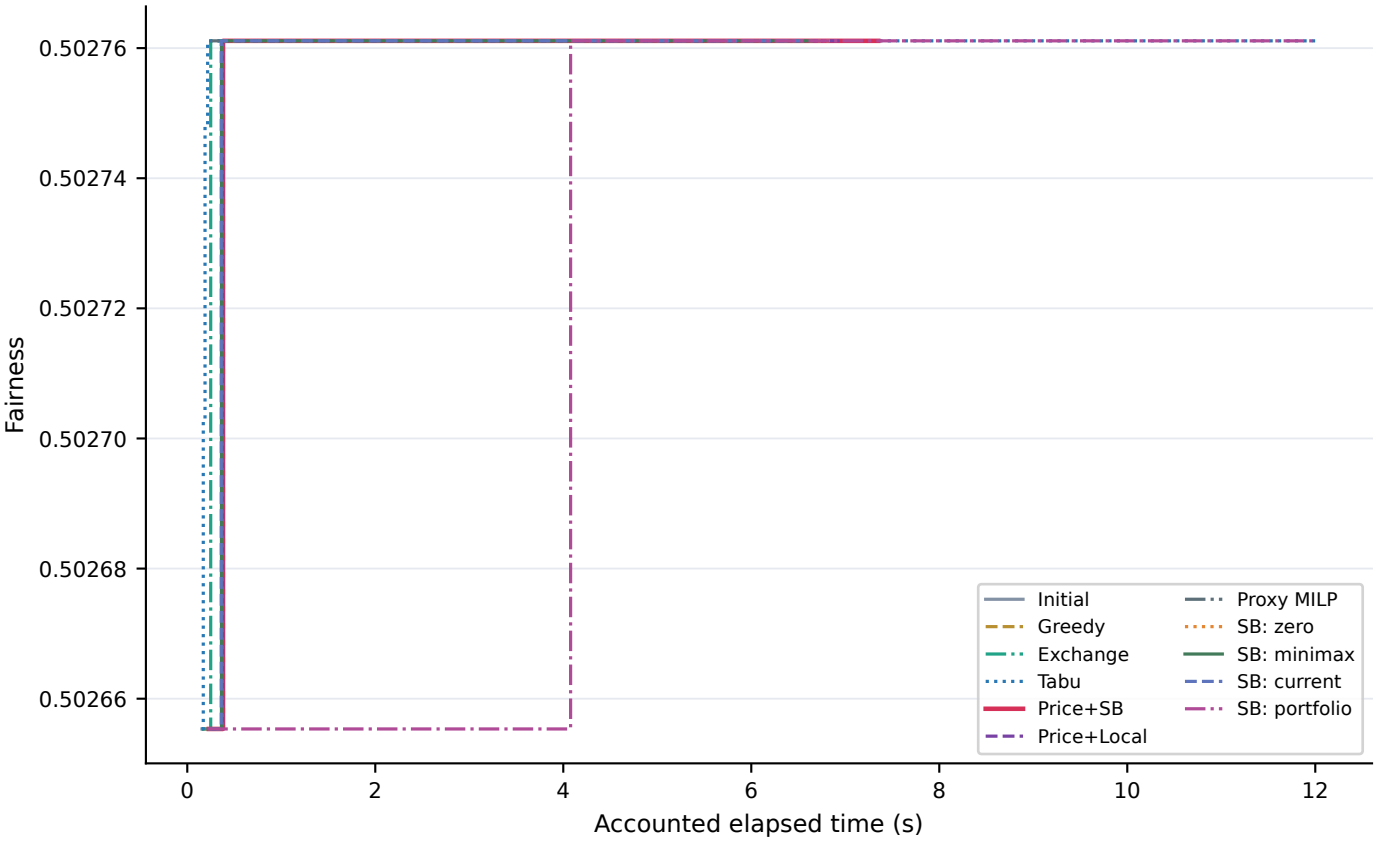


Saved throughput: phase 2

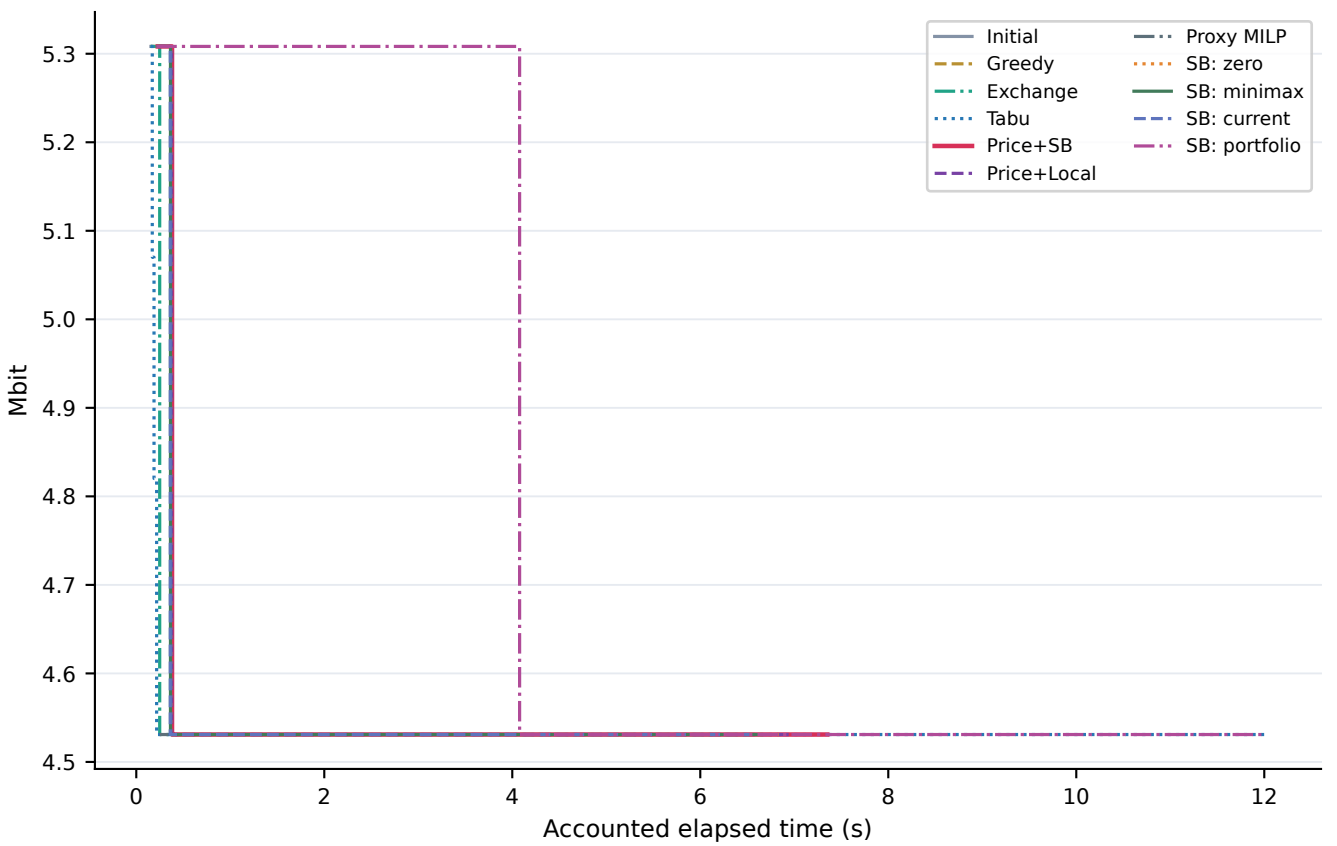


small | first requested seed 20261105 | recorded checkpoints; no solver-internal interpolation

Original fairness versus accounted time

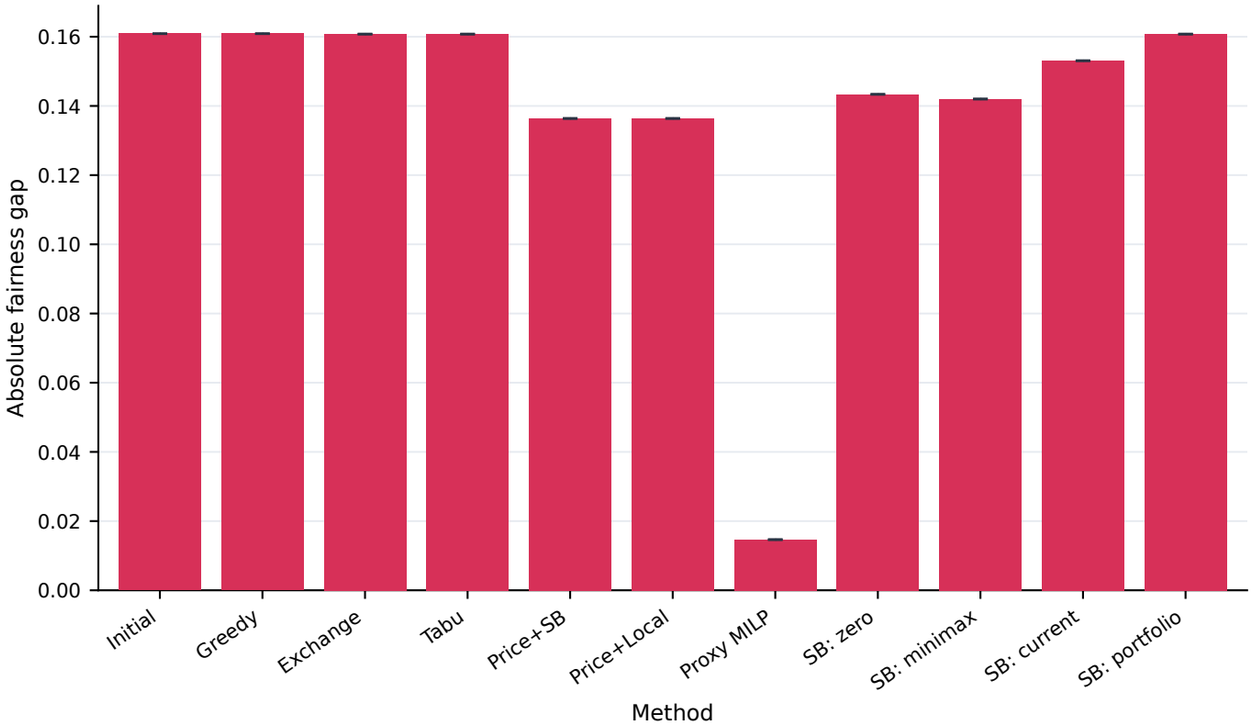


Original throughput versus accounted time

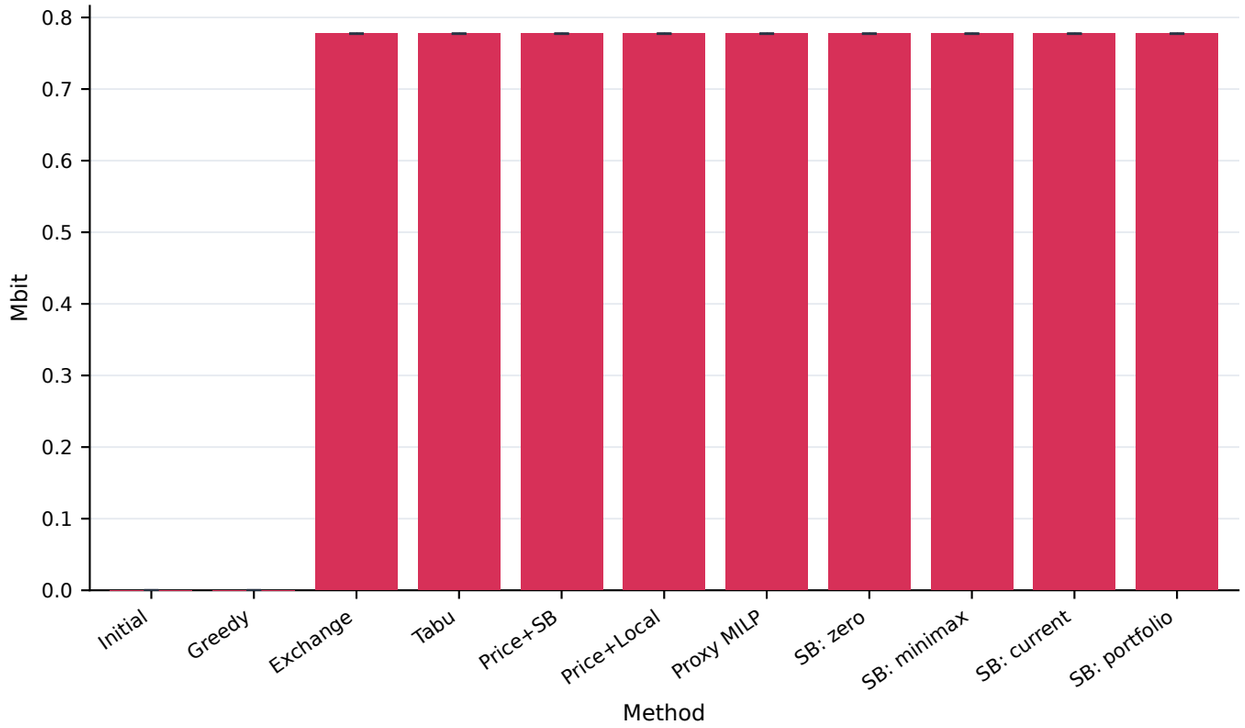


small | K=8, M=3, L=8 | median and min-max across 3 seeds

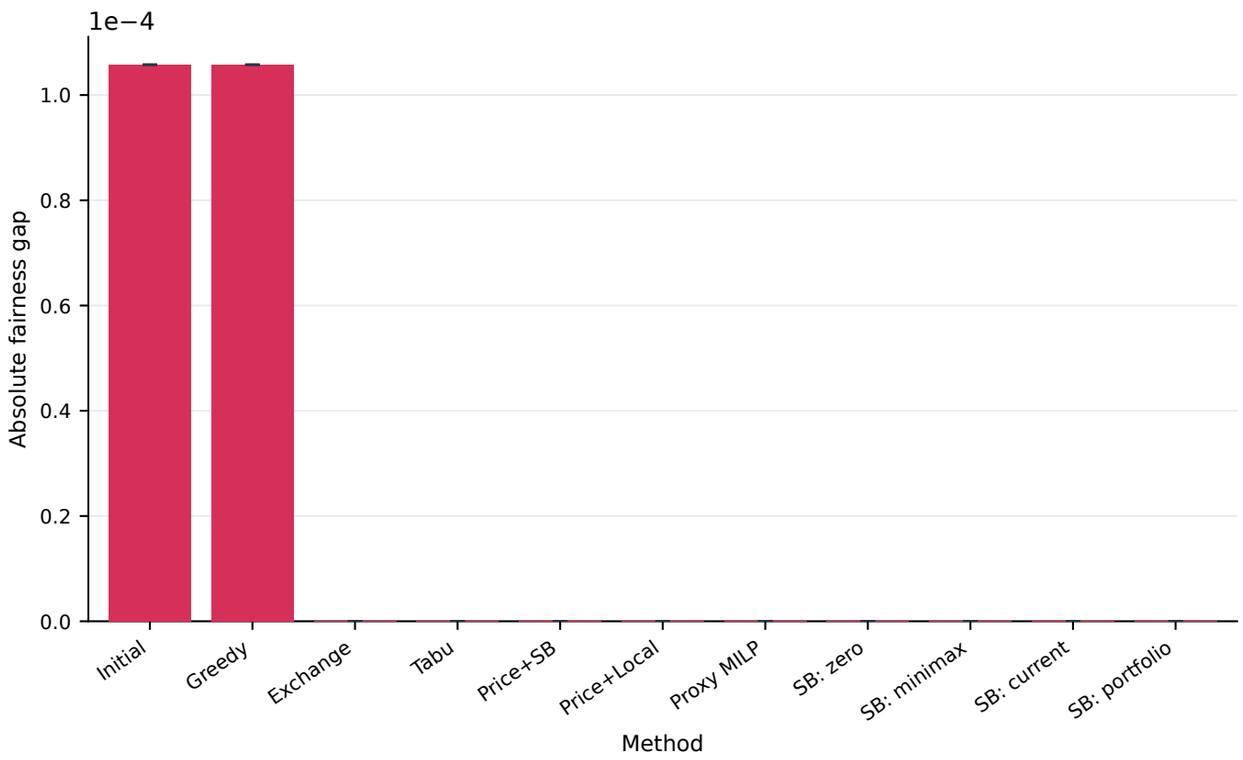
Valid original fairness bound minus achieved value



Valid throughput gap at each method own floor

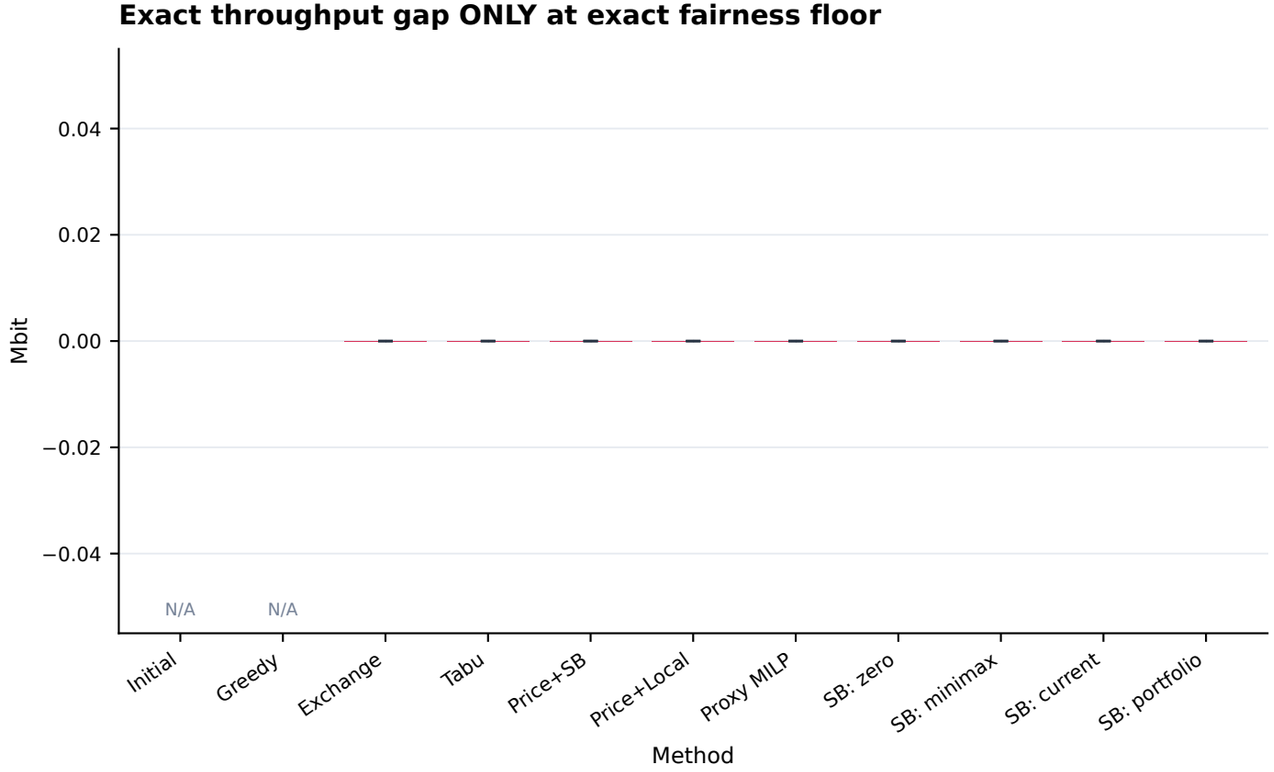


Gap to exhaustive fairness optimum



Only a completed exhaustive search supplies an exact reference.

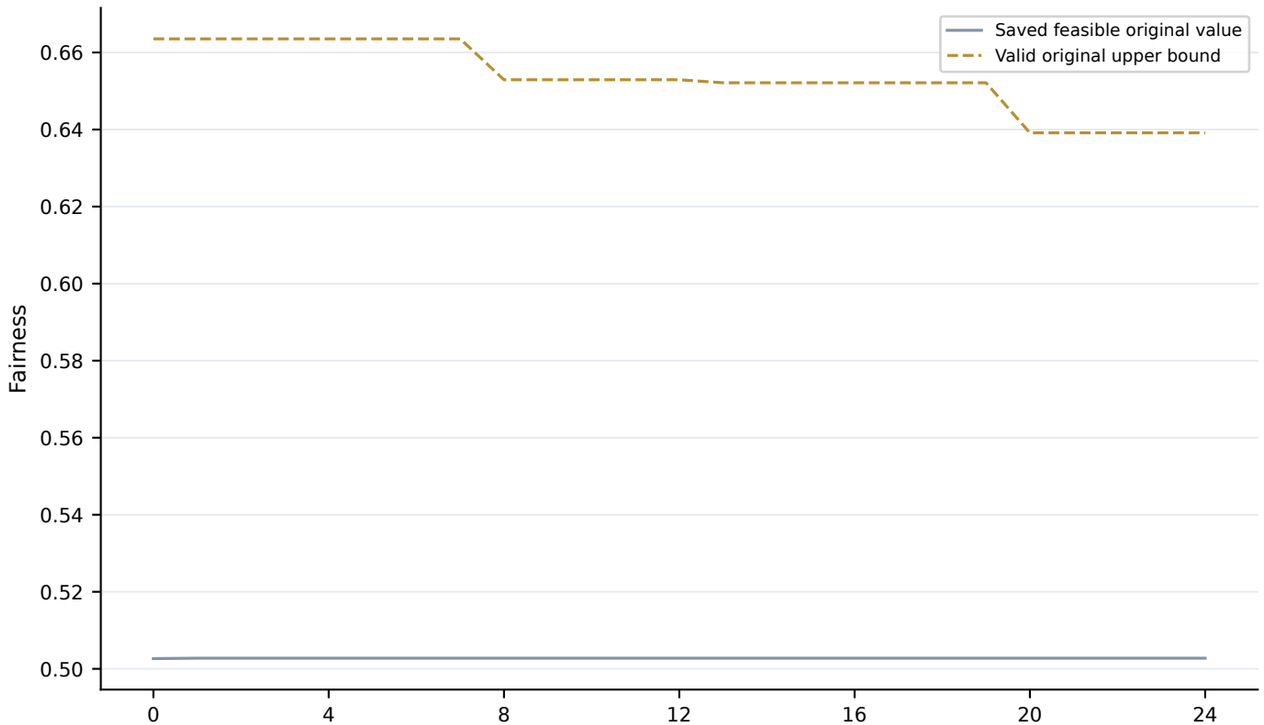
Exact throughput gap ONLY at exact fairness floor



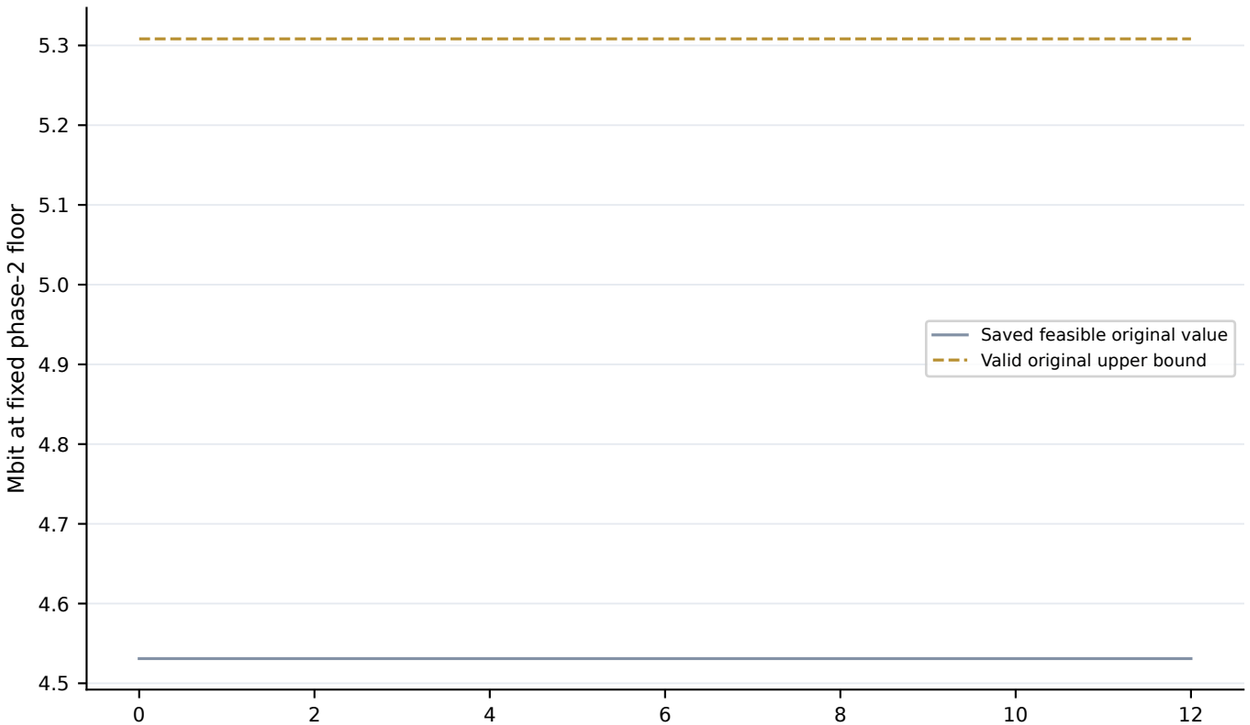
Different fairness floors are not comparable; these entries remain missing.

small | first requested seed 20261105

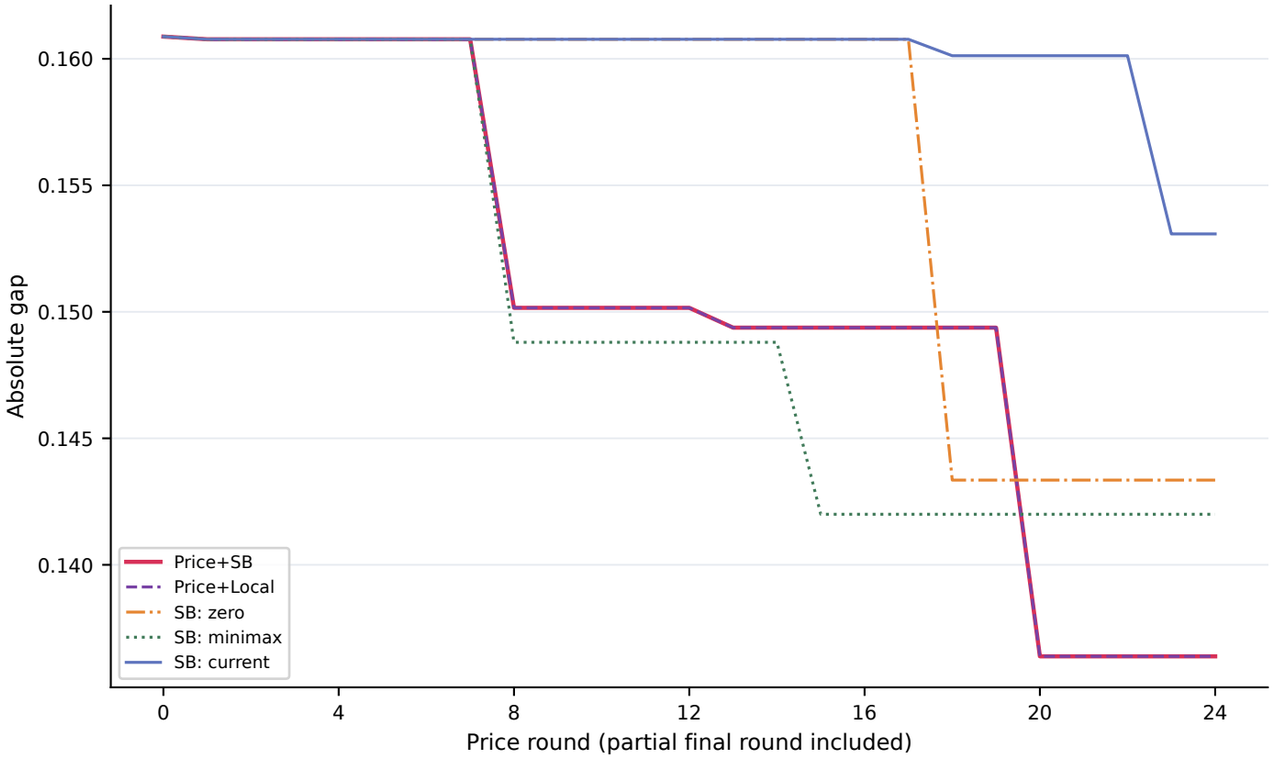
Price+SB valid original bounds: phase 1



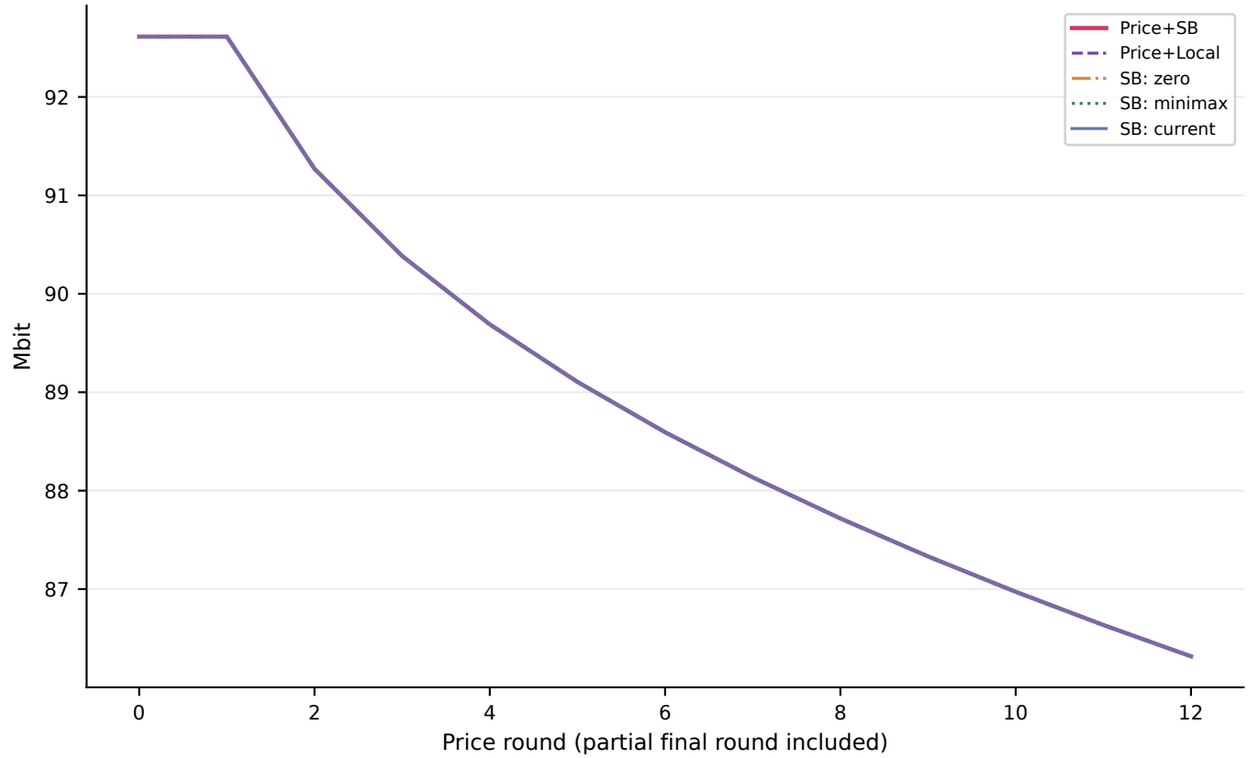
Price+SB valid original bounds: phase 2



Valid fairness gap during price search

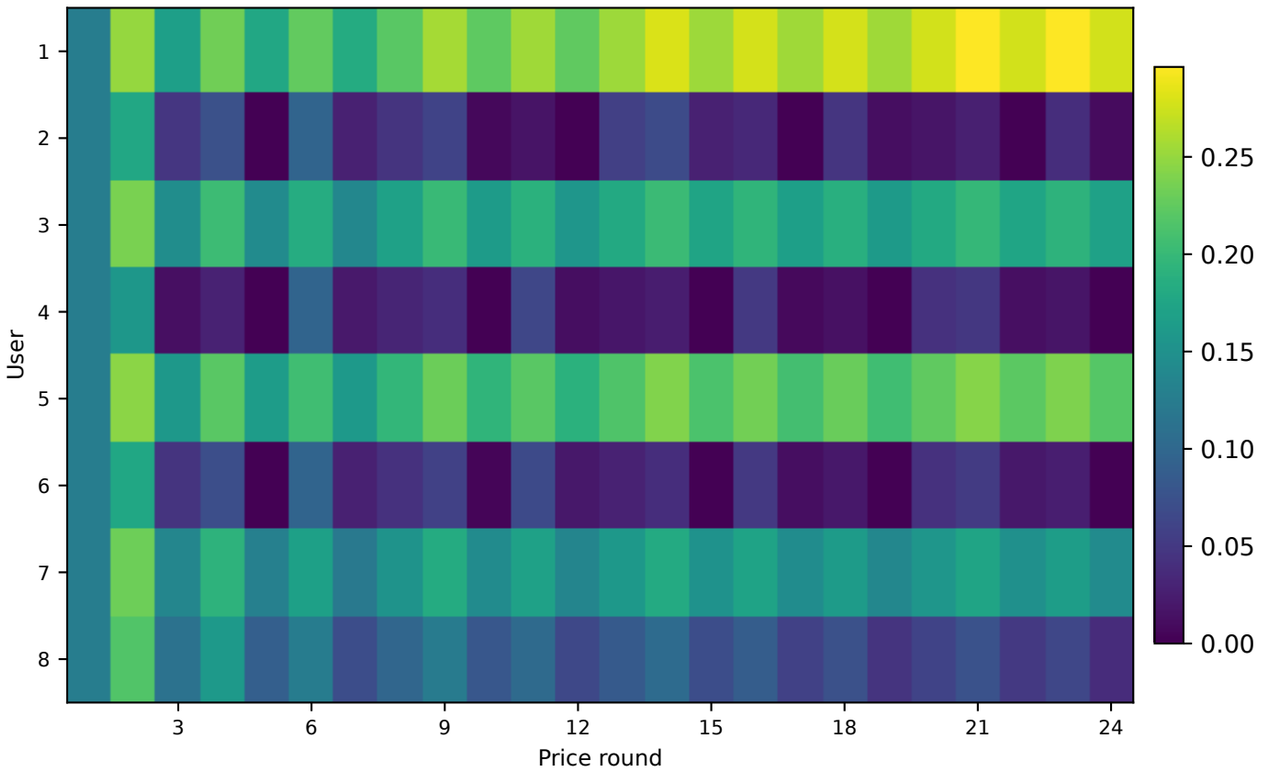


Valid throughput gap during price search

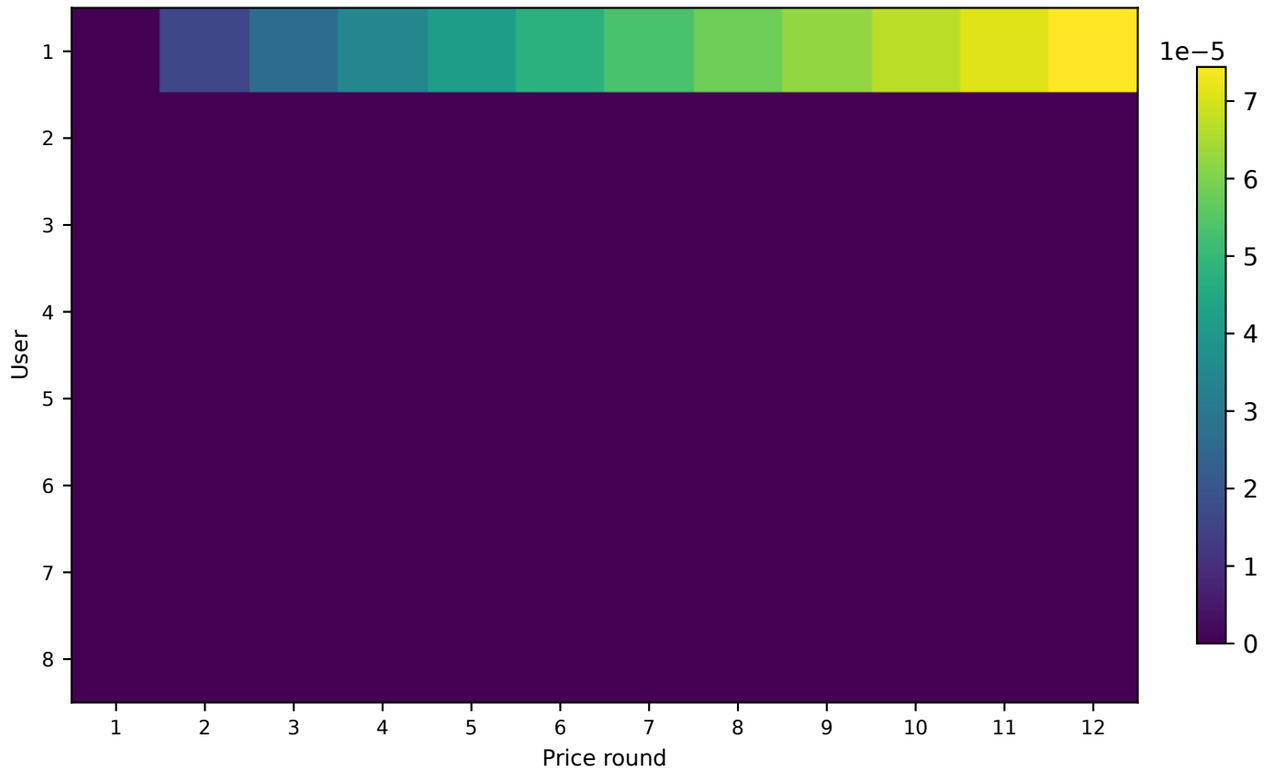


small | first requested seed 20261105 | Price+SB

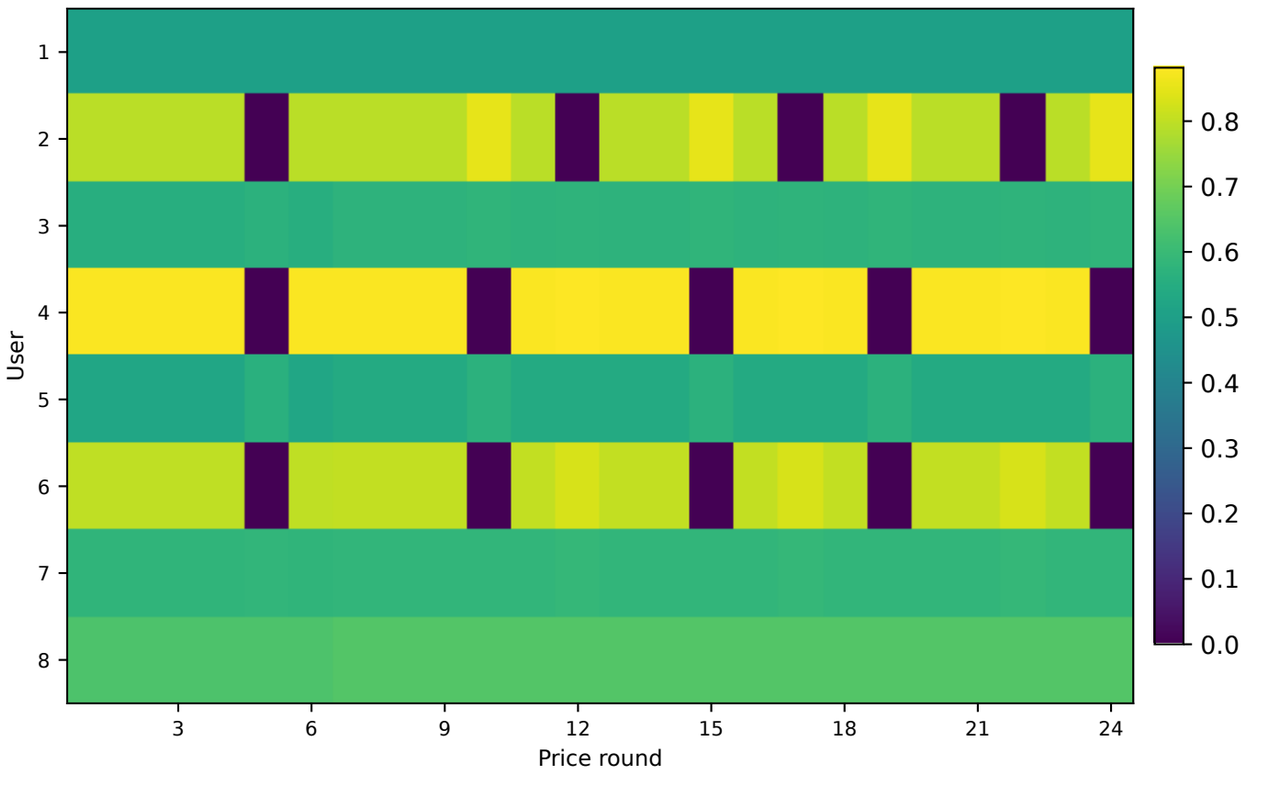
Phase 1 nonnegative user prices



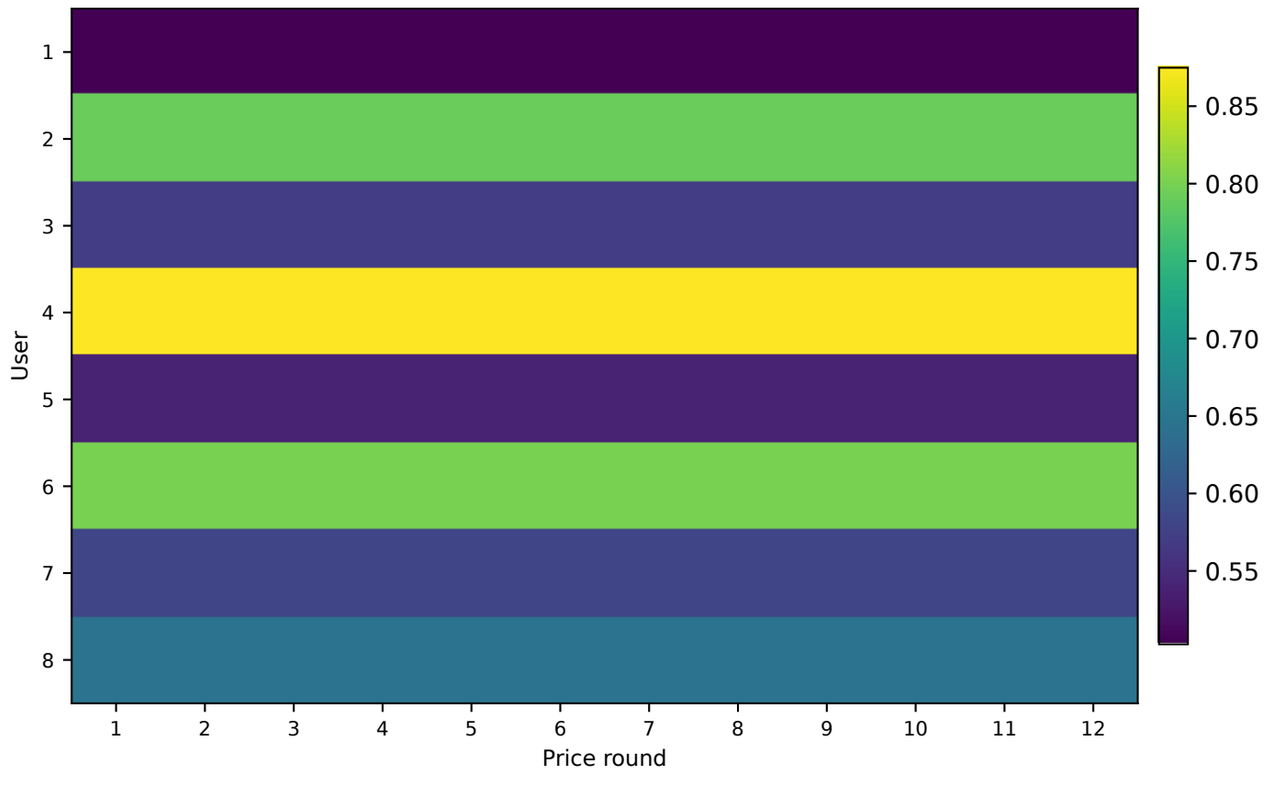
Phase 2 nonnegative user prices



Phase 1 lower service / target (unclipped)

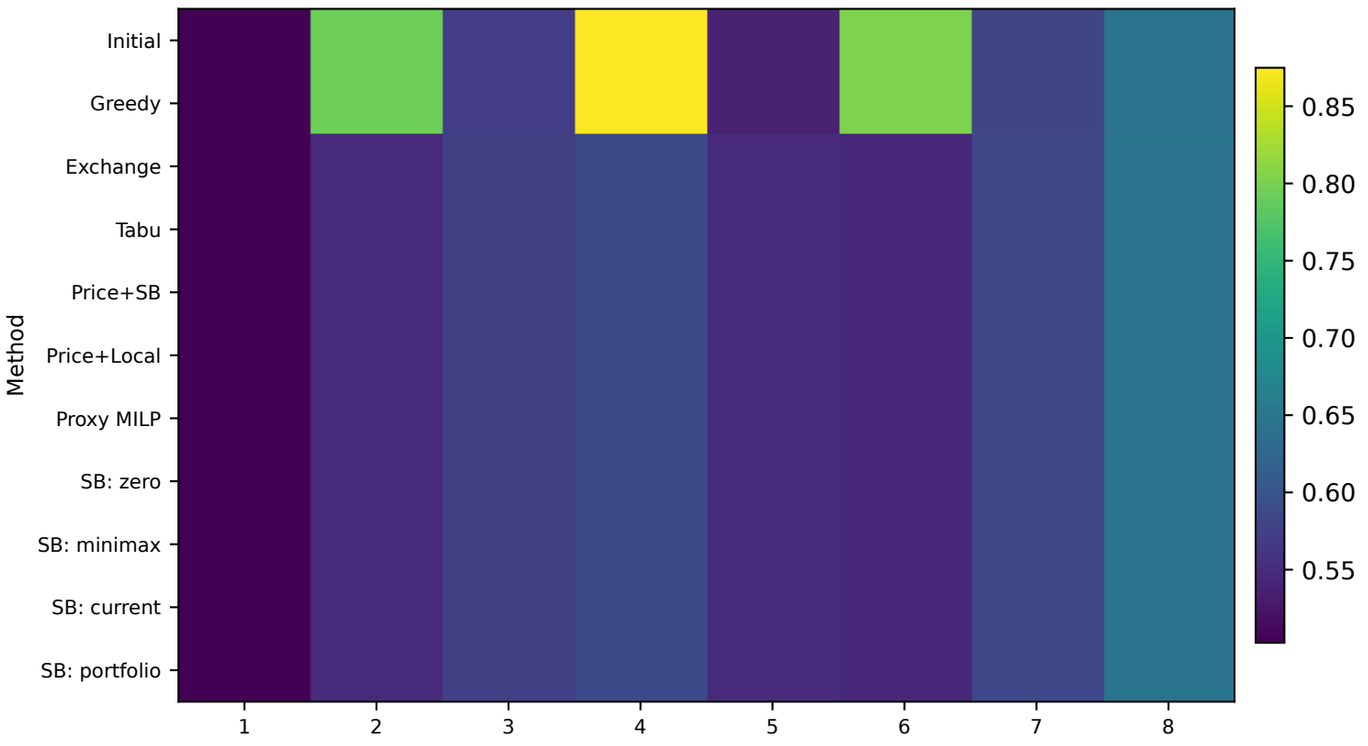


Phase 2 lower service / target (unclipped)

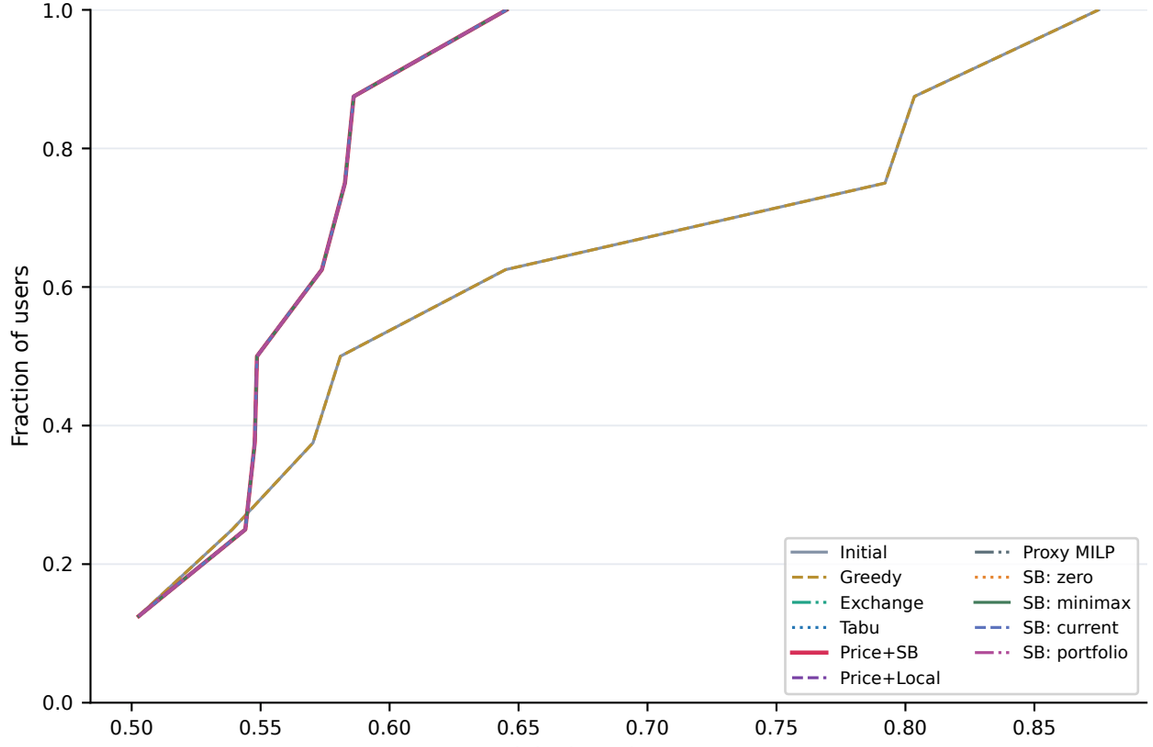


small | first requested seed 20261105

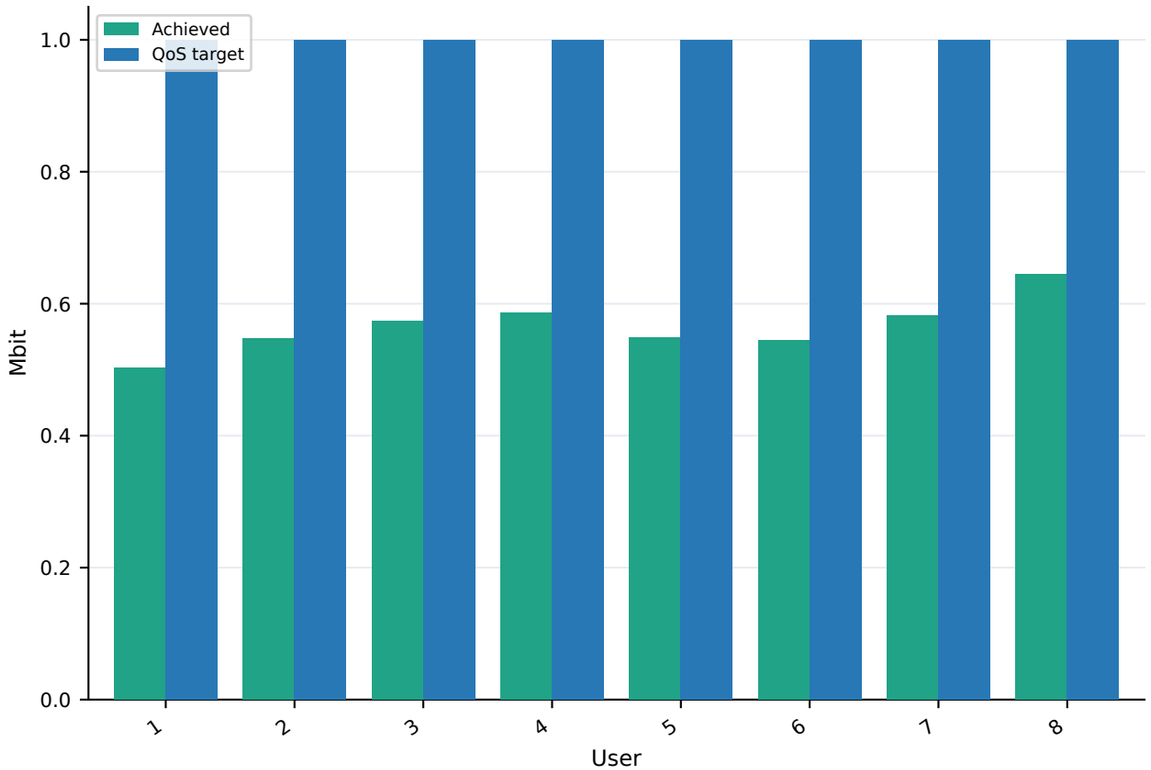
Every user original demand ratio



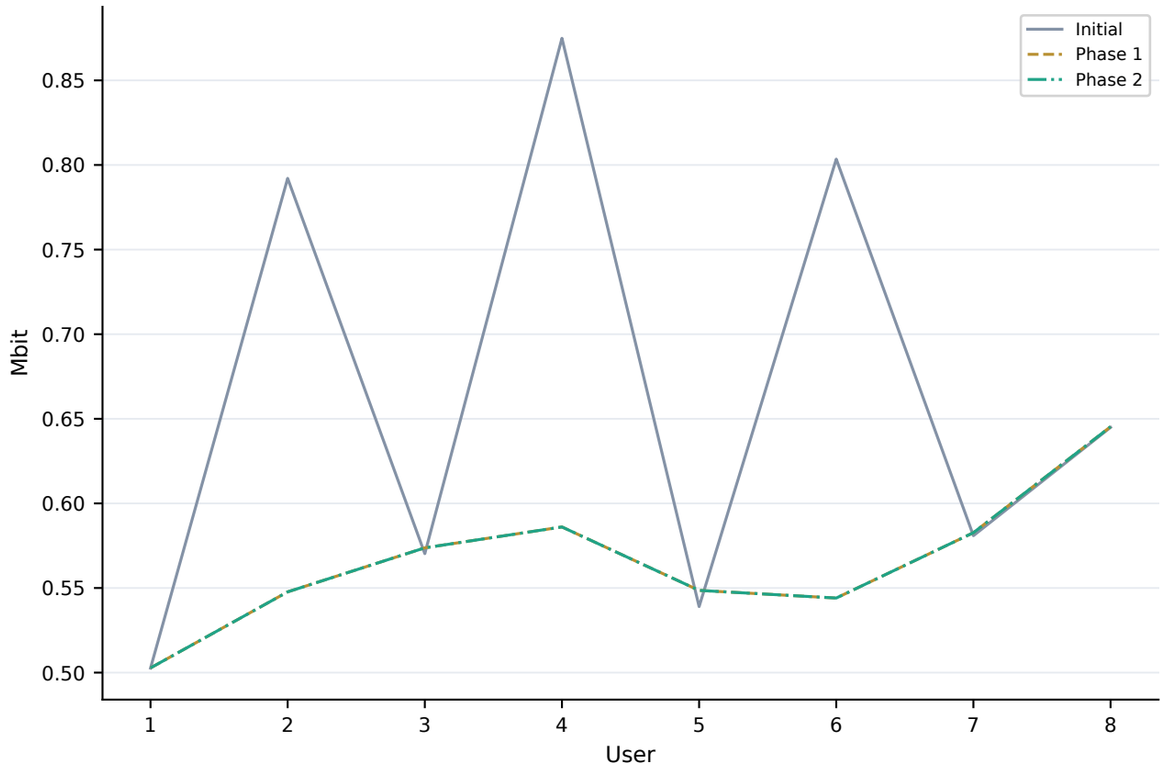
Empirical CDF of user demand ratios



Price+SB service and full QoS target



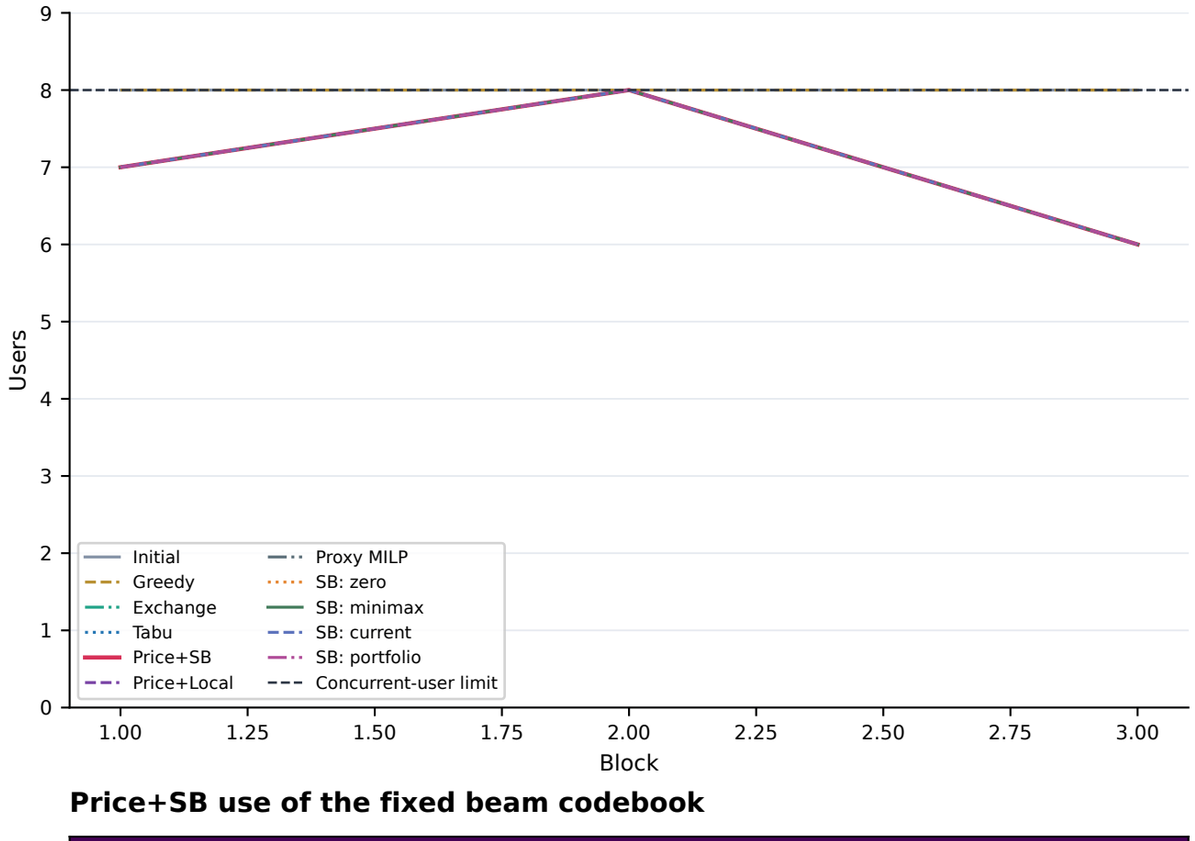
Price+SB per-user service across phases



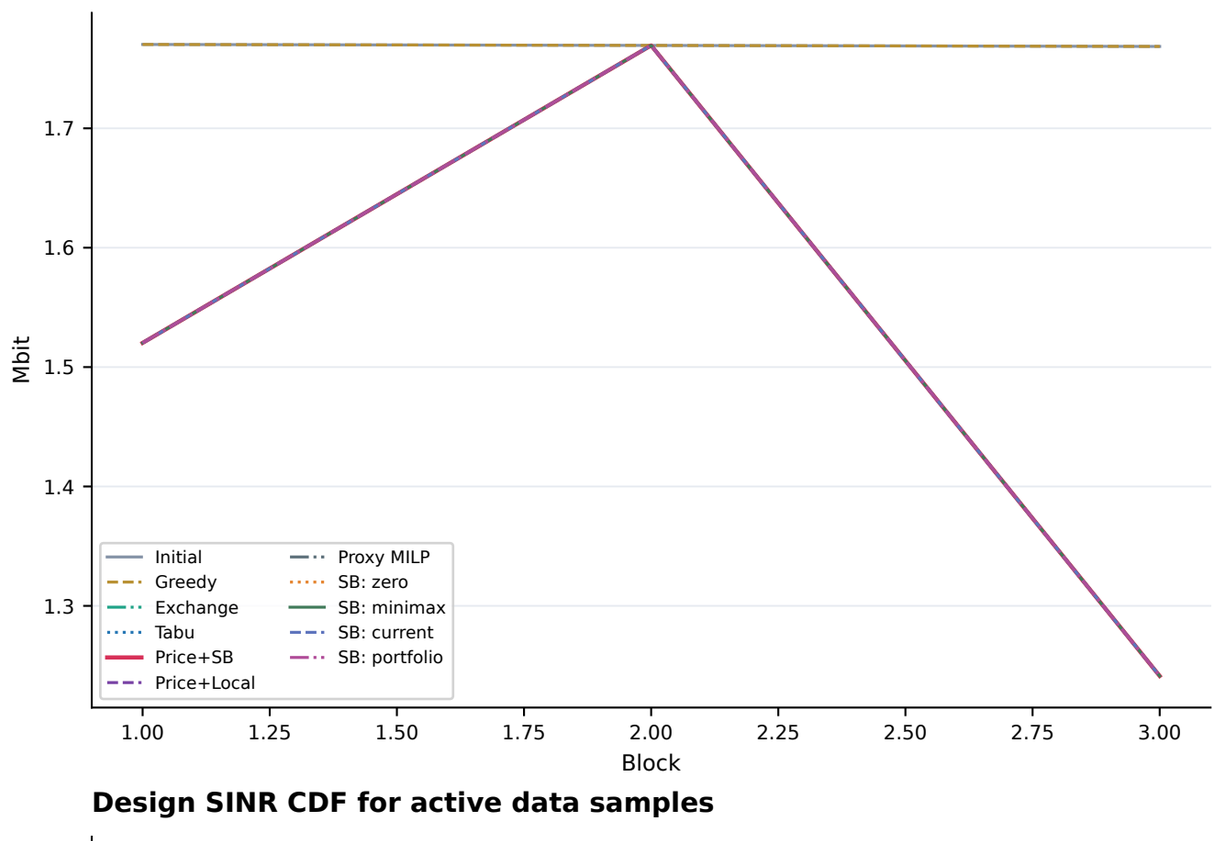
small | first requested seed 20261105 | 1=selected, 0=idle; one decision per fixed block

small | first requested seed 20261105

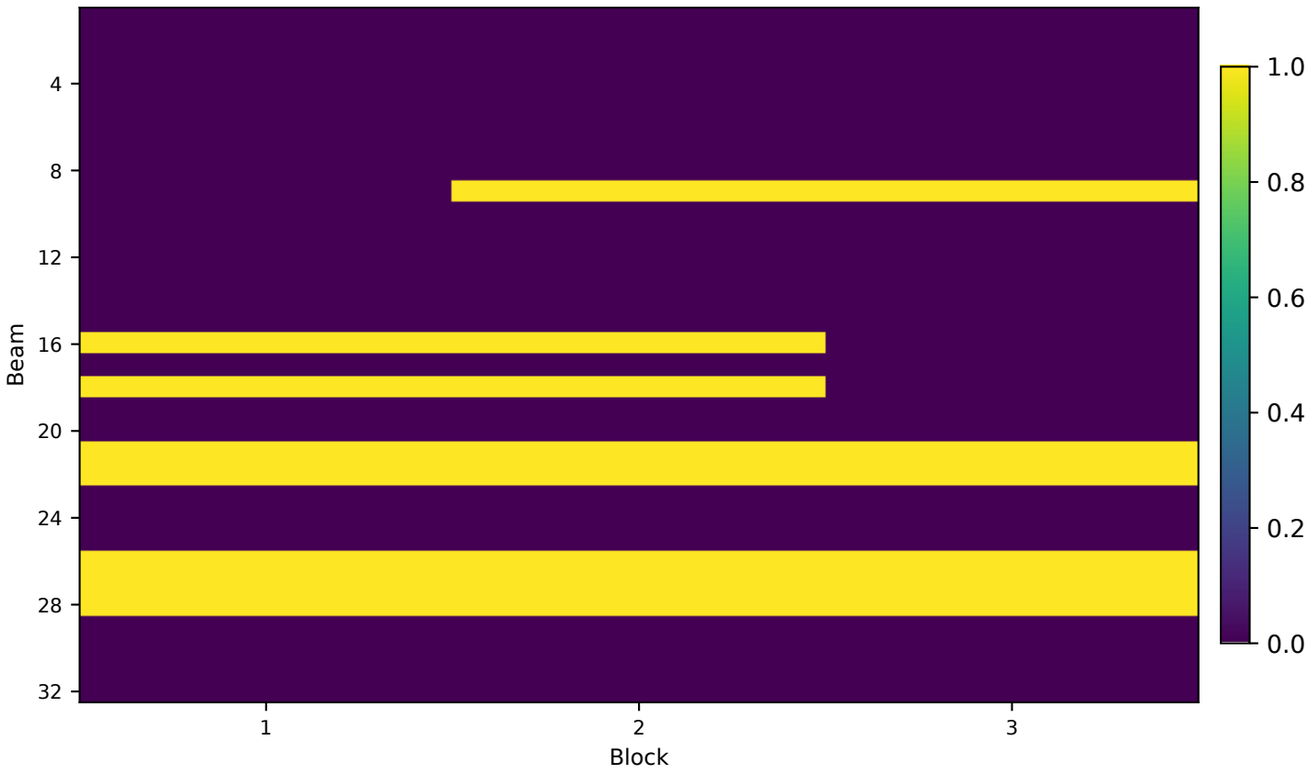
Concurrent users per fixed block



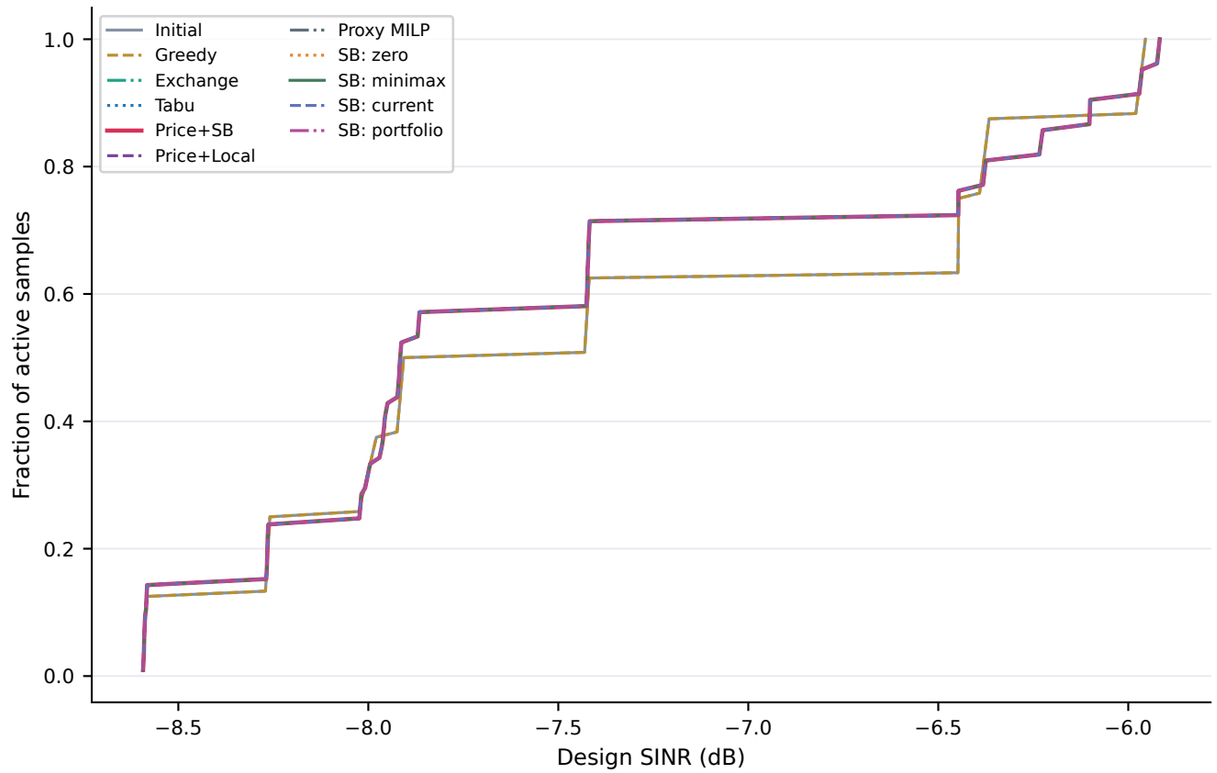
Net service per fixed block



Price+SB use of the fixed beam codebook

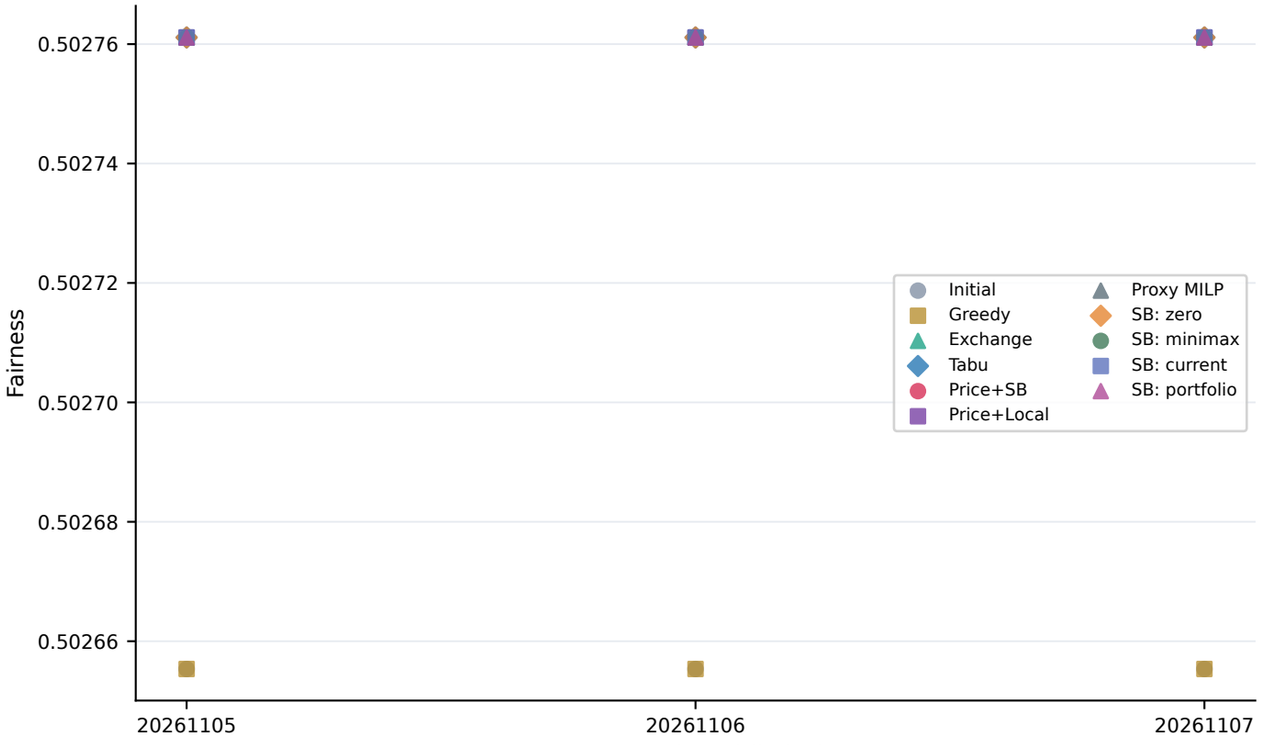


Design SINR CDF for active data samples

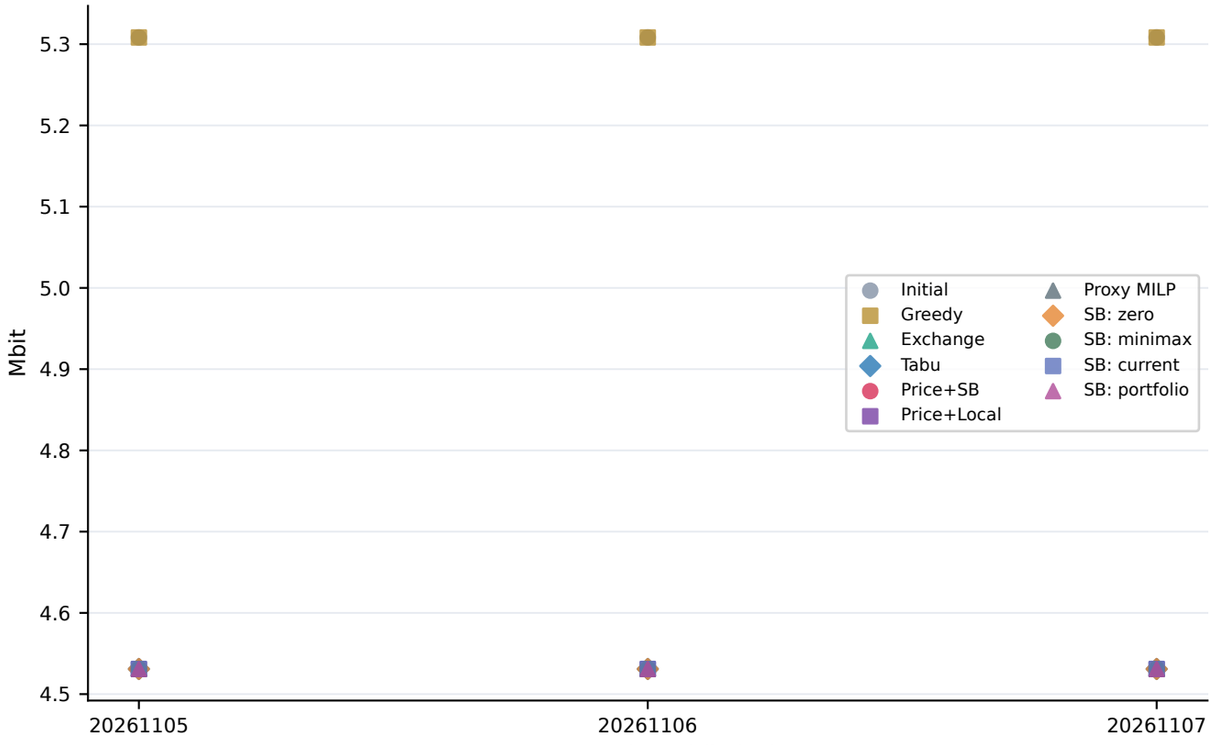


small | K=8, M=3, L=8 | median and min-max across 3 seeds

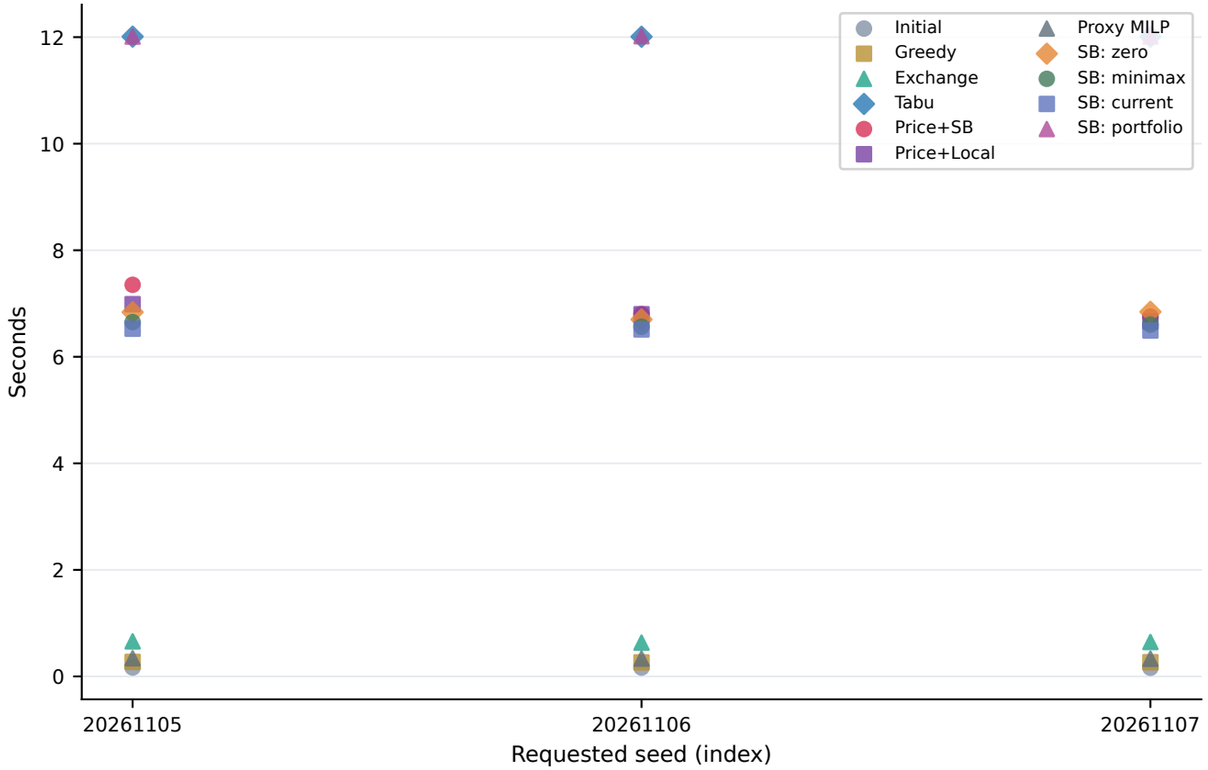
Fairness for every seed



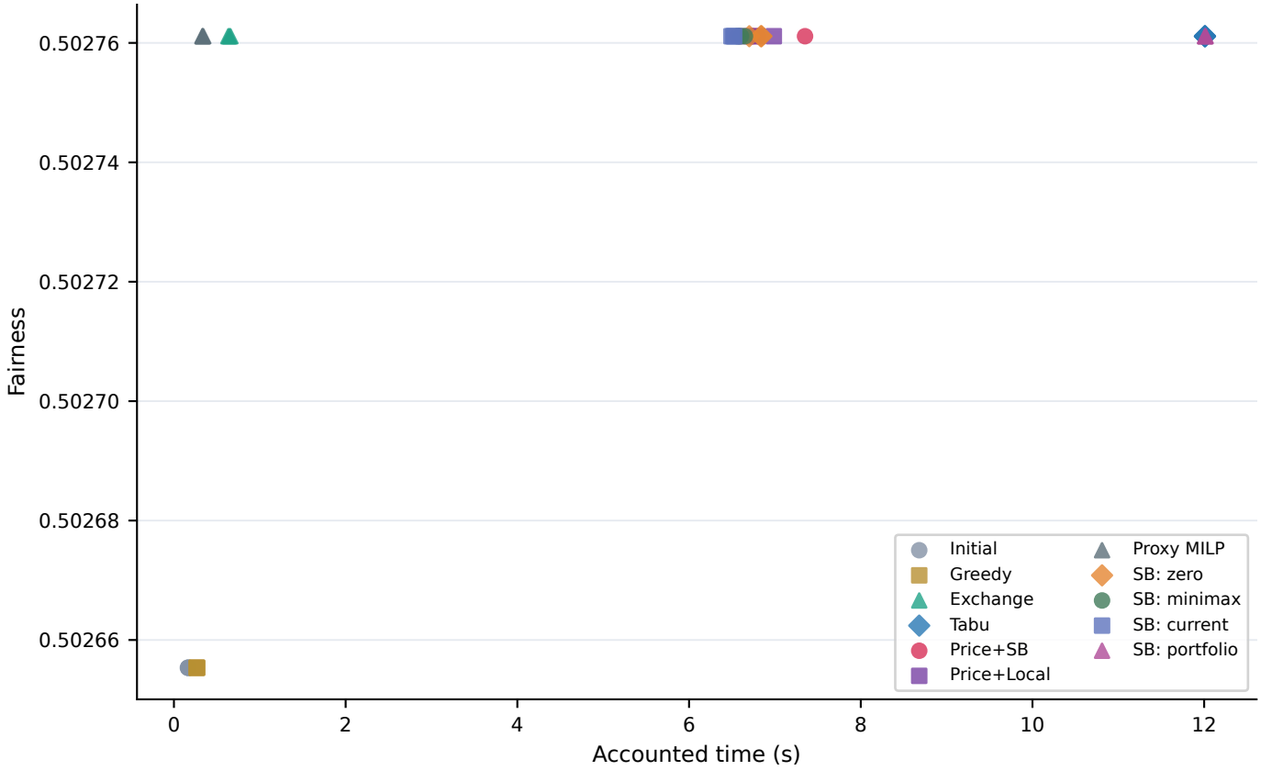
Throughput for every seed



Time for every seed

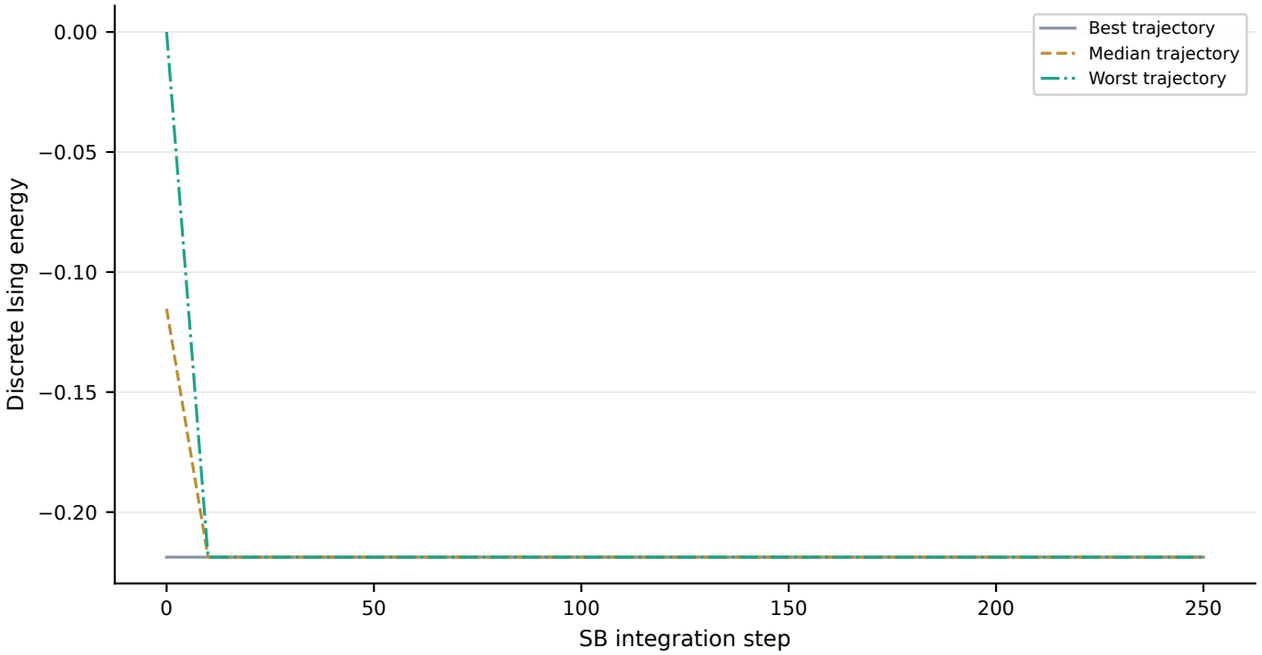


Quality versus time (every completed run)



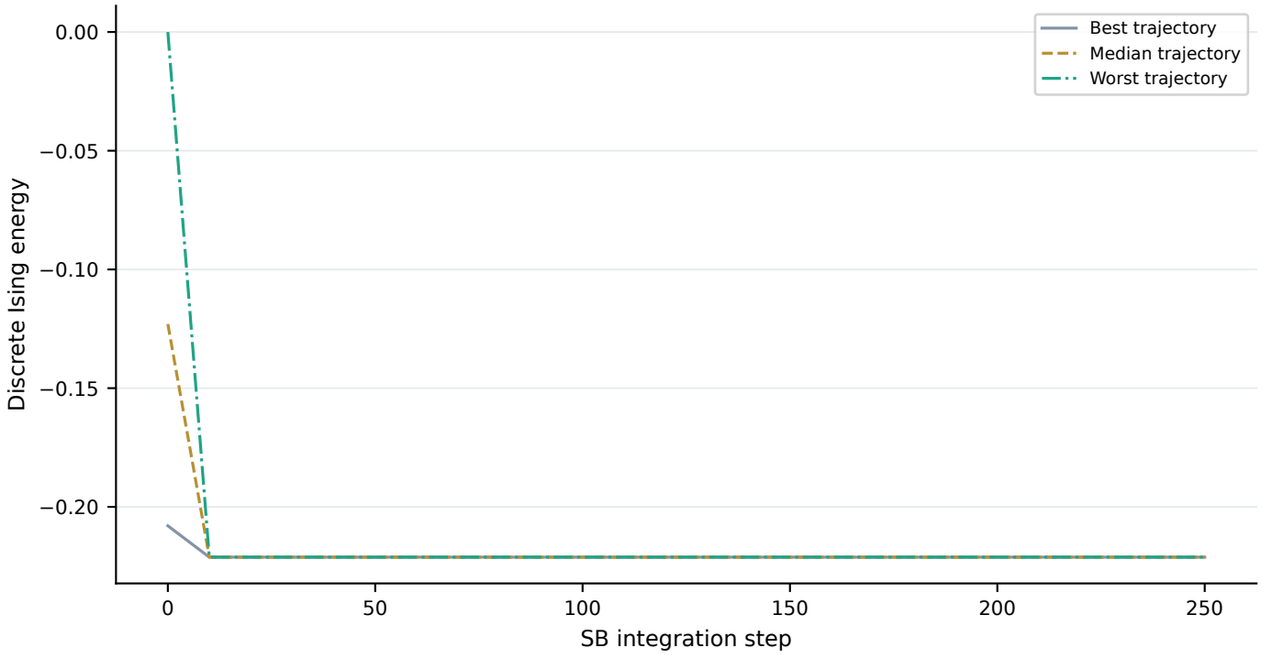
small | first requested seed 20261105 | first Price+SB Ising call per phase; before hard repair

Phase 1: resource 1, price round 1, 16 trajectories



Different phases solve different Ising problems; their energy values are not comparable.

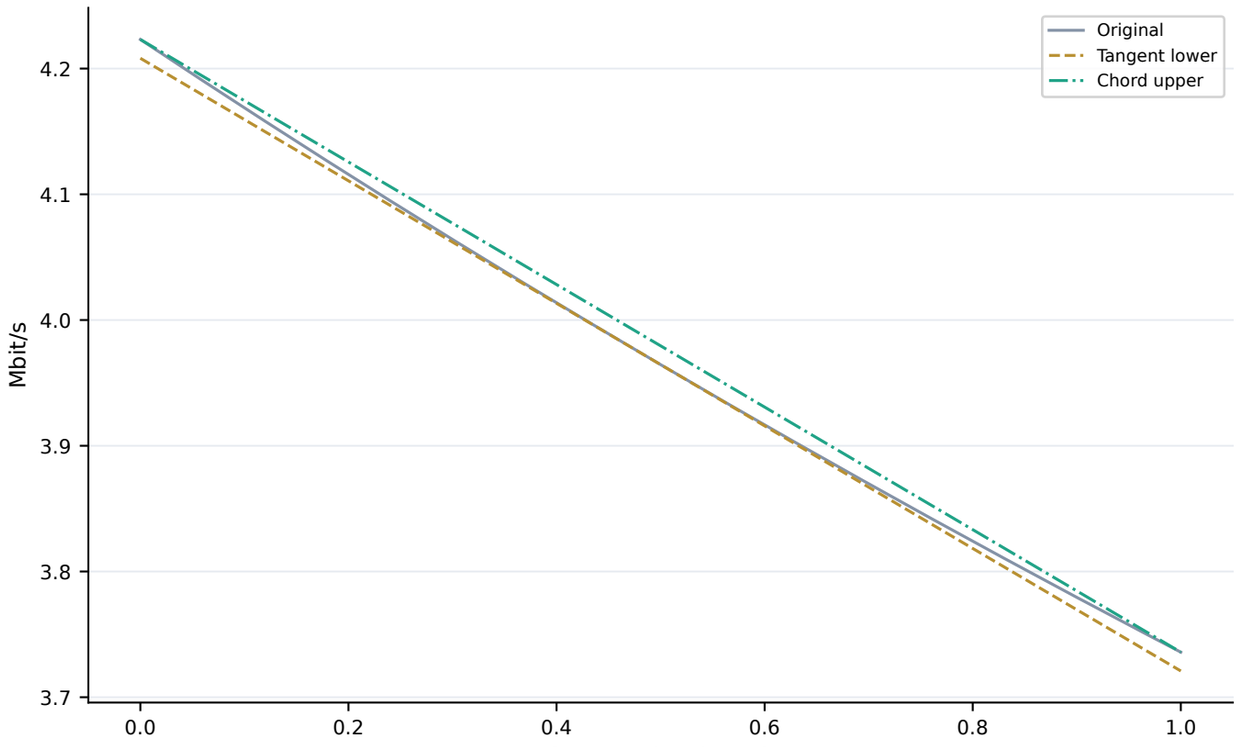
Phase 2: resource 1, price round 1, 16 trajectories



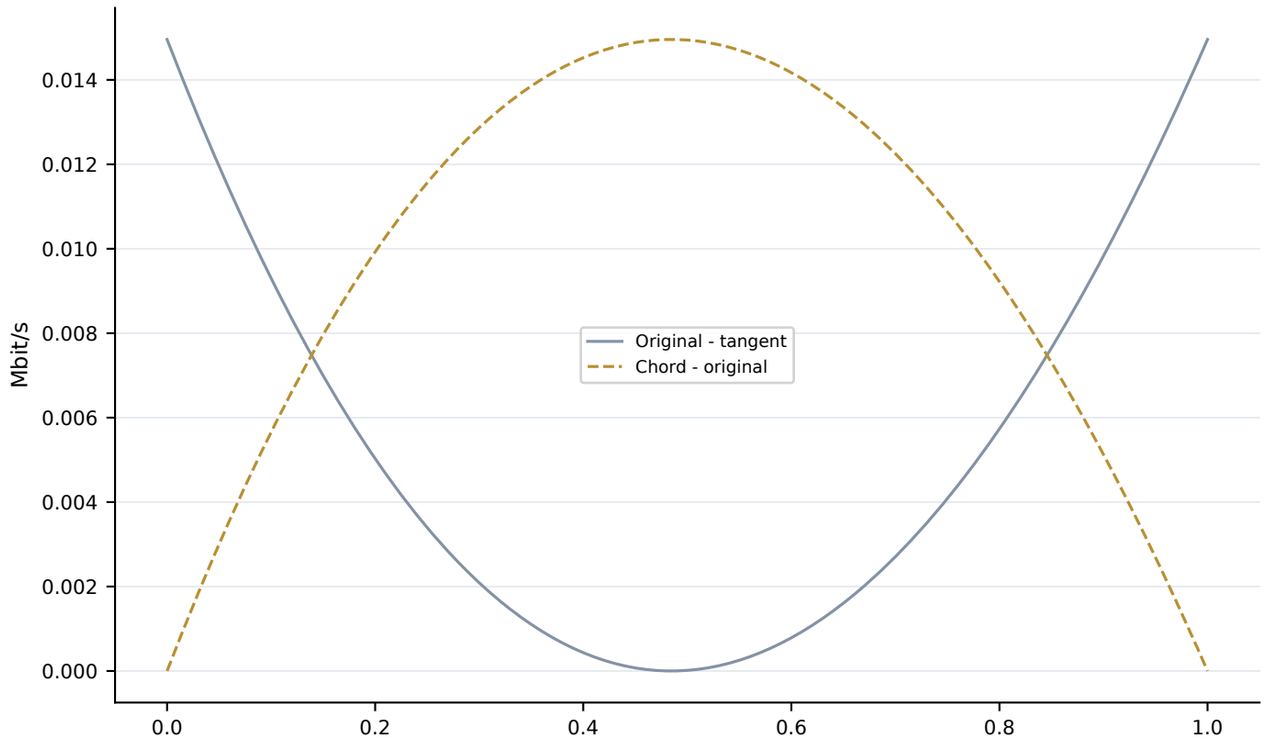
Different phases solve different Ising problems; their energy values are not comparable.

small | first requested seed 20261105

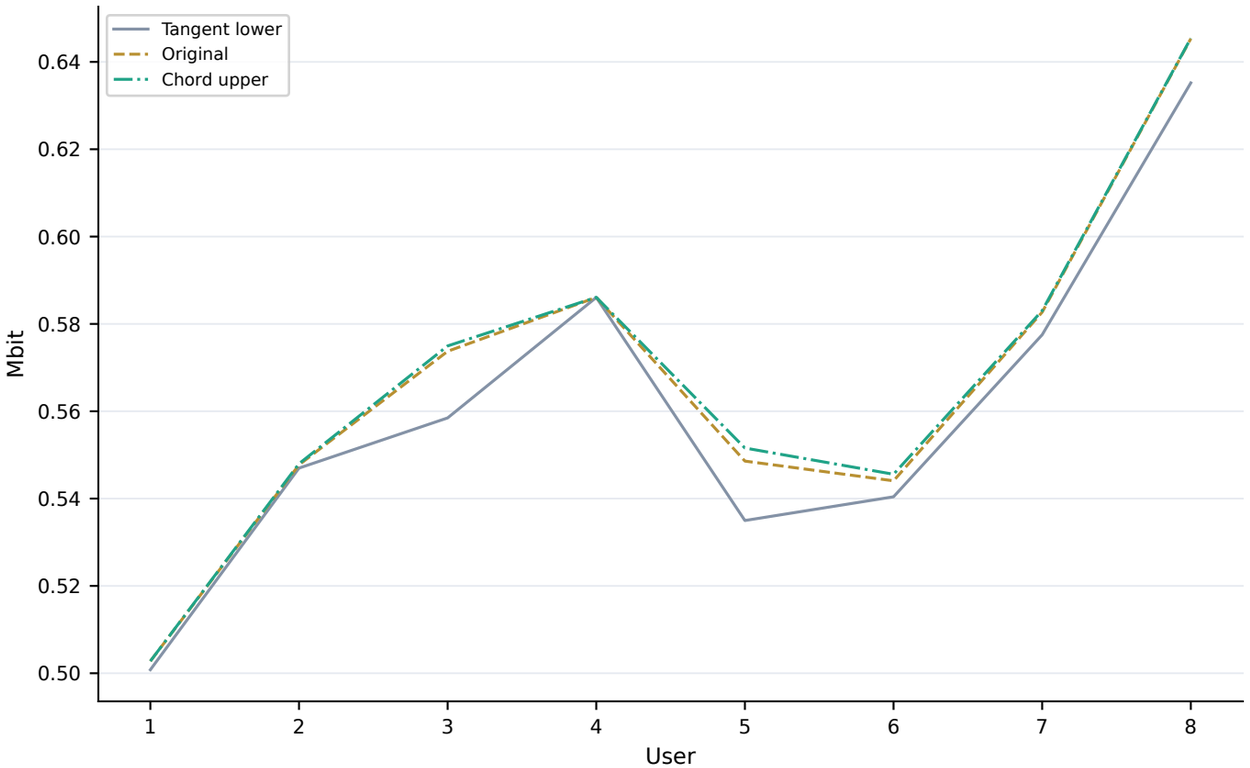
Rate bounds, first nonzero interval: user 1, sample 1



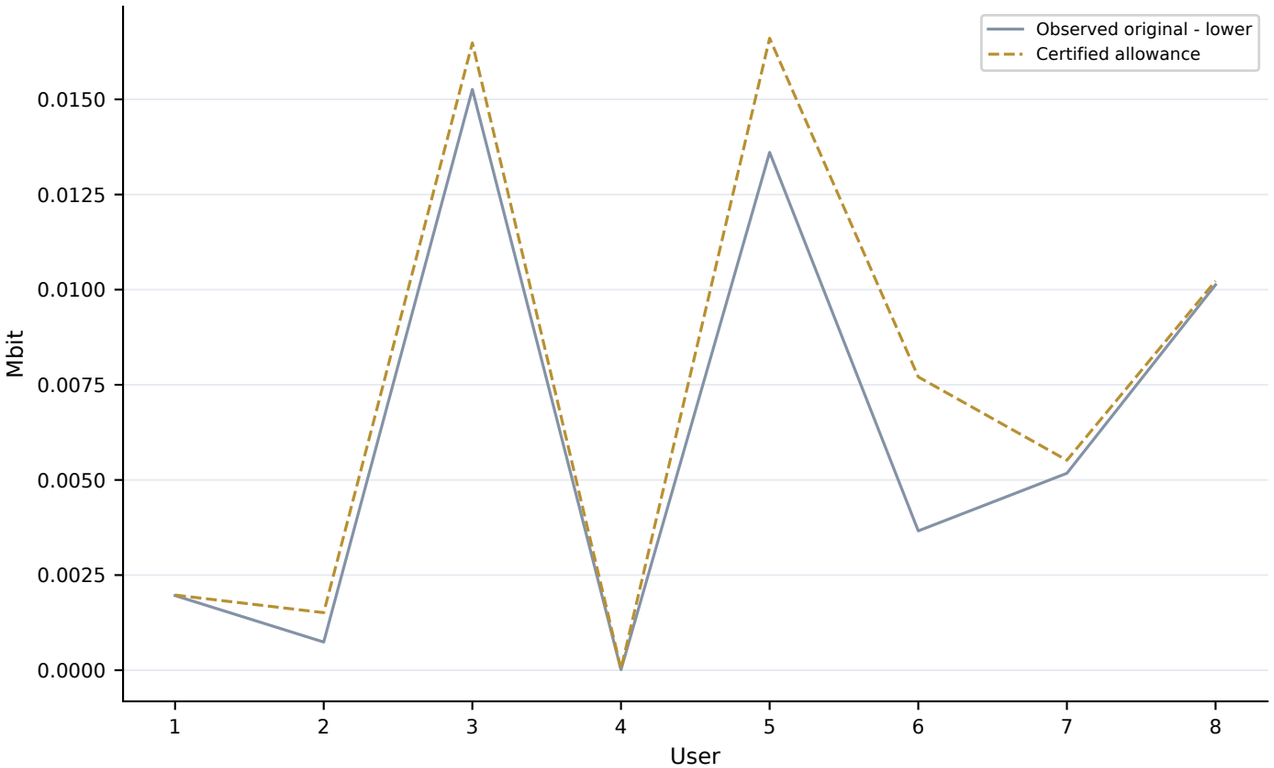
Rate approximation error over legal interval



Price+SB final schedule: fixed minimax service bounds

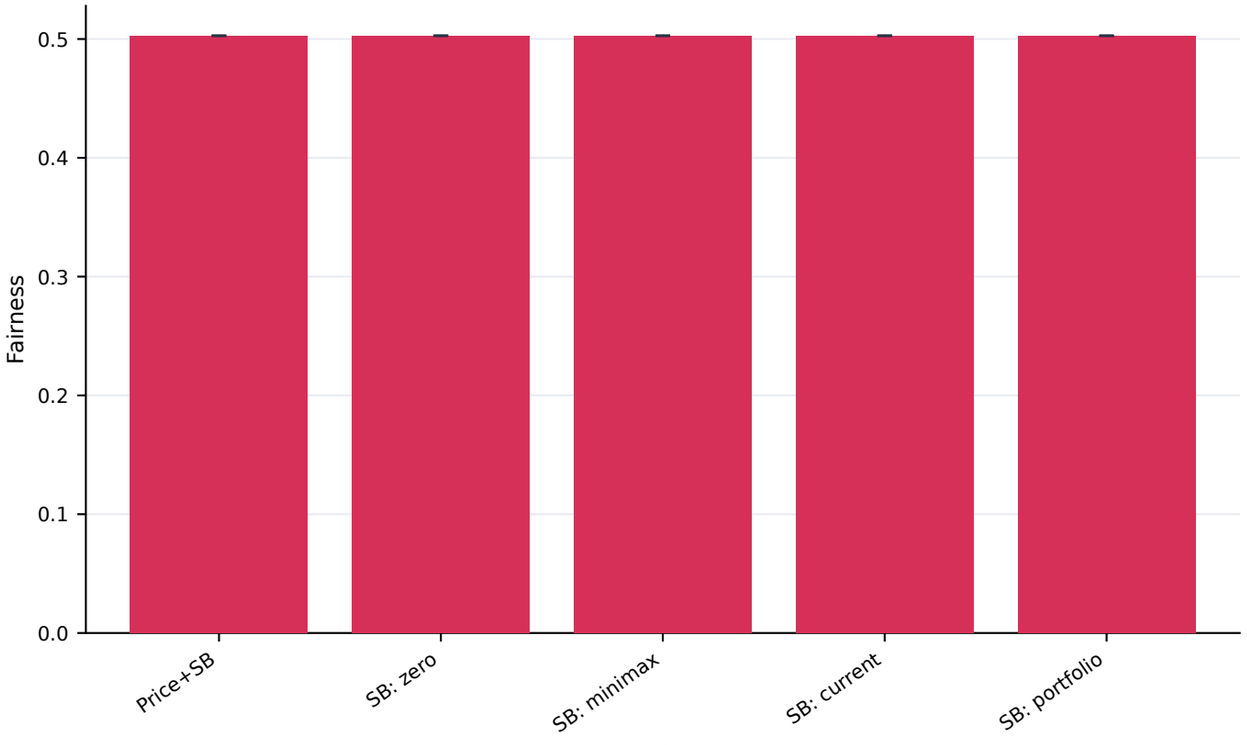


Observed lower error and certified error allowance

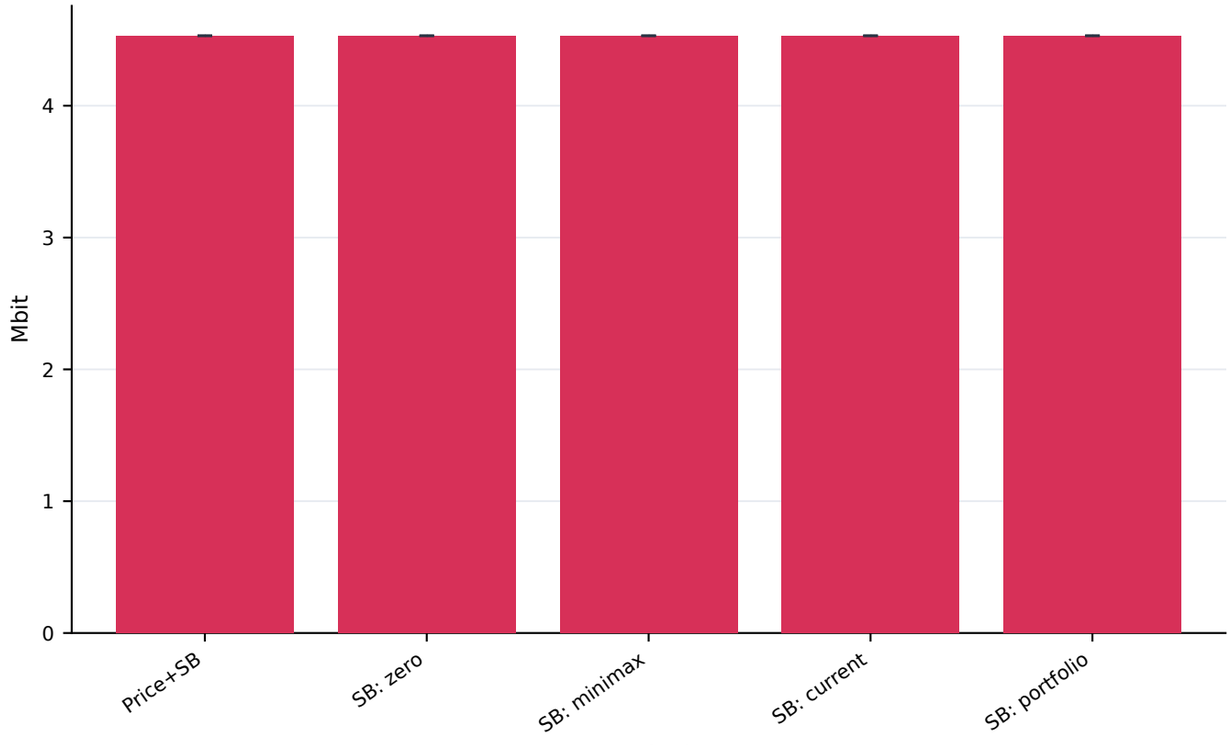


small | K=8, M=3, L=8 | median and min-max across 3 seeds

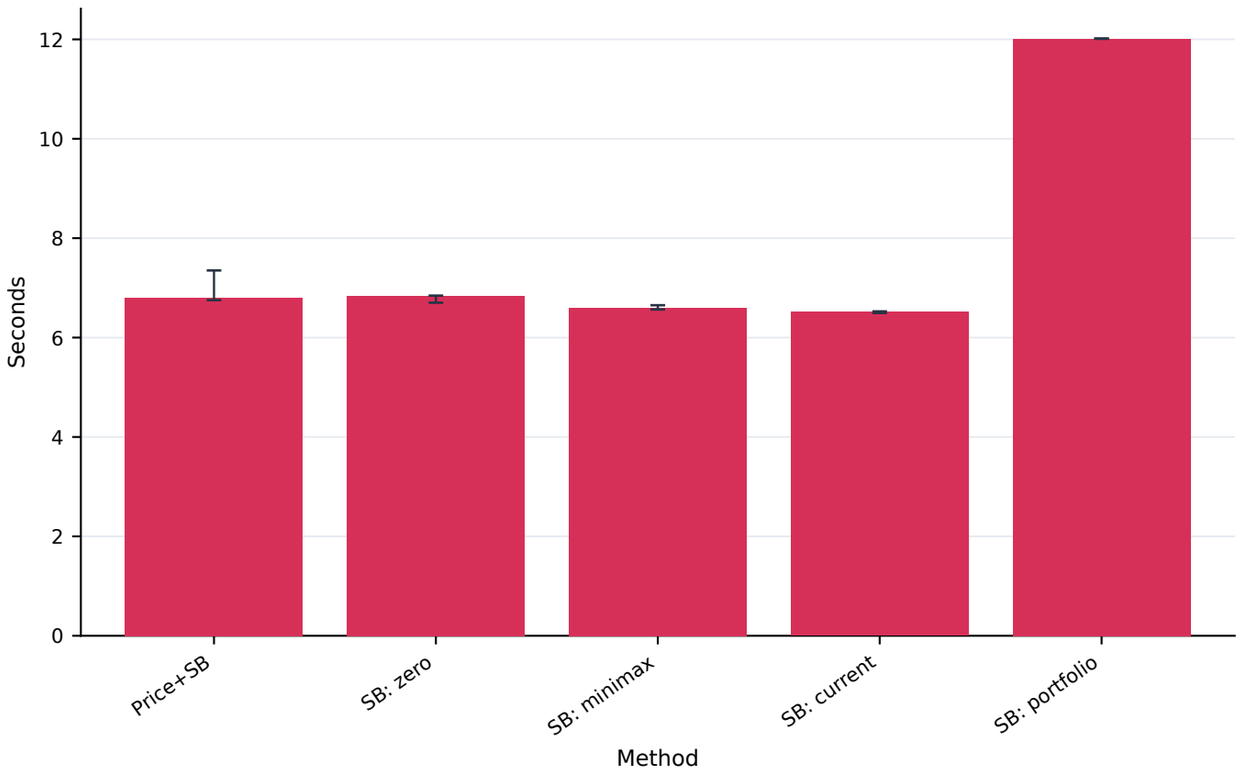
Reference-point ablation: original fairness



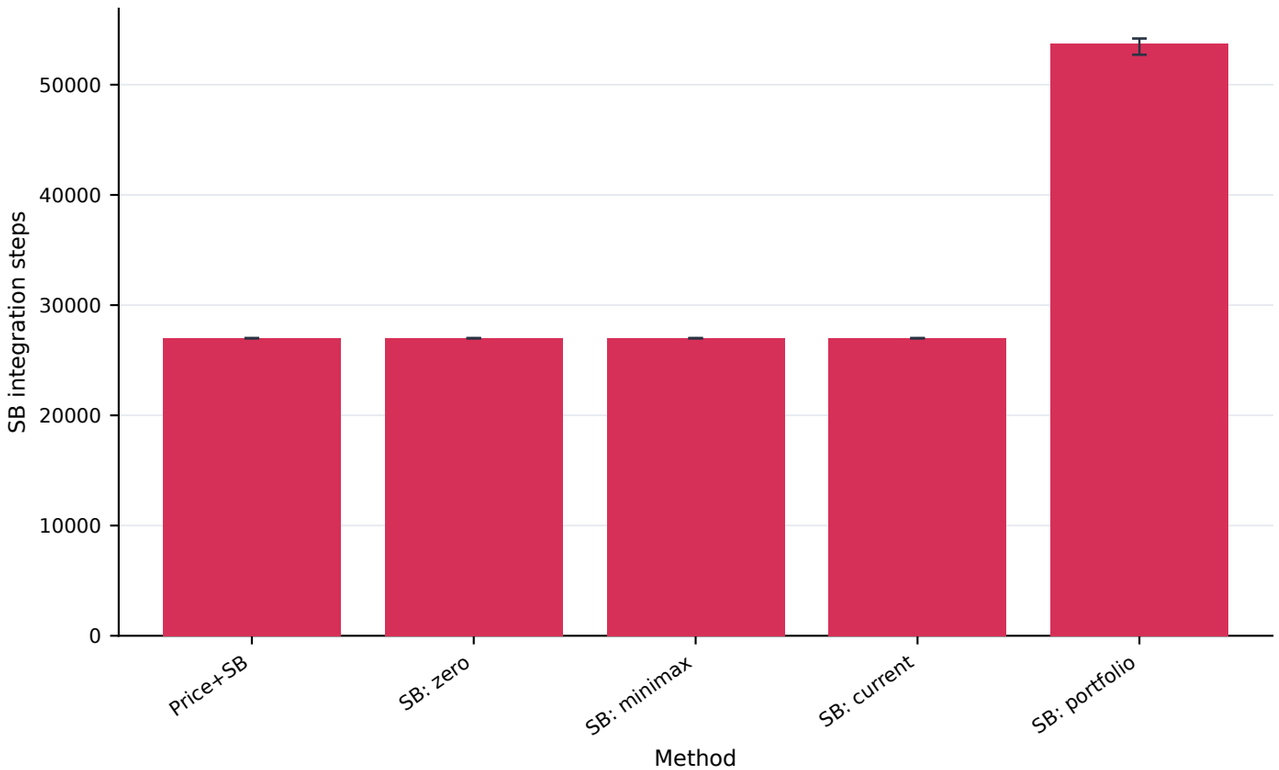
Reference-point ablation: original throughput

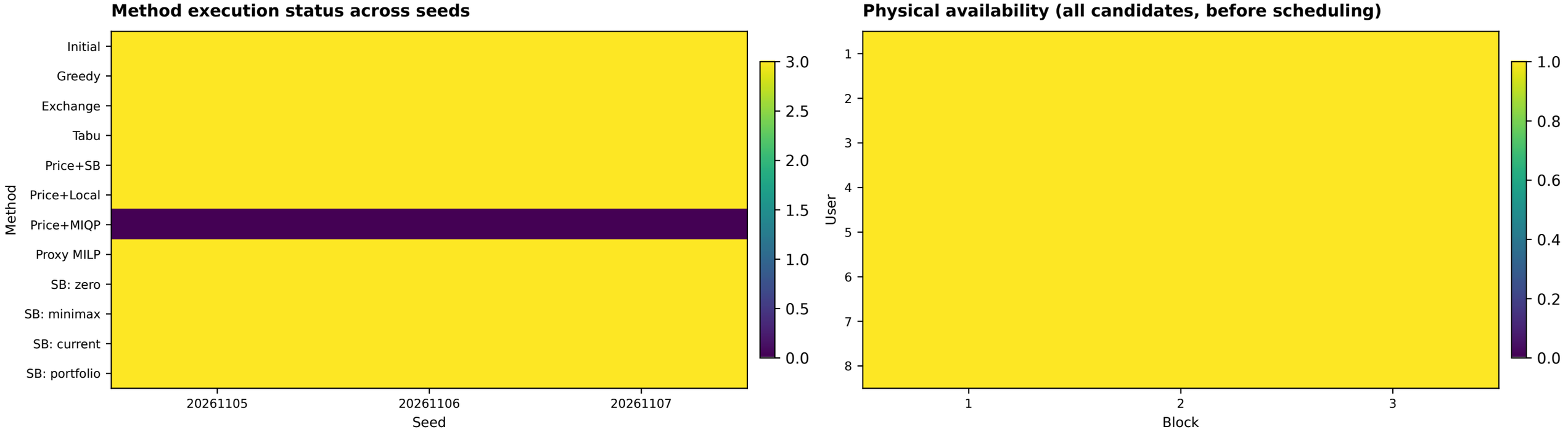


Reference-point ablation: actual accounted time



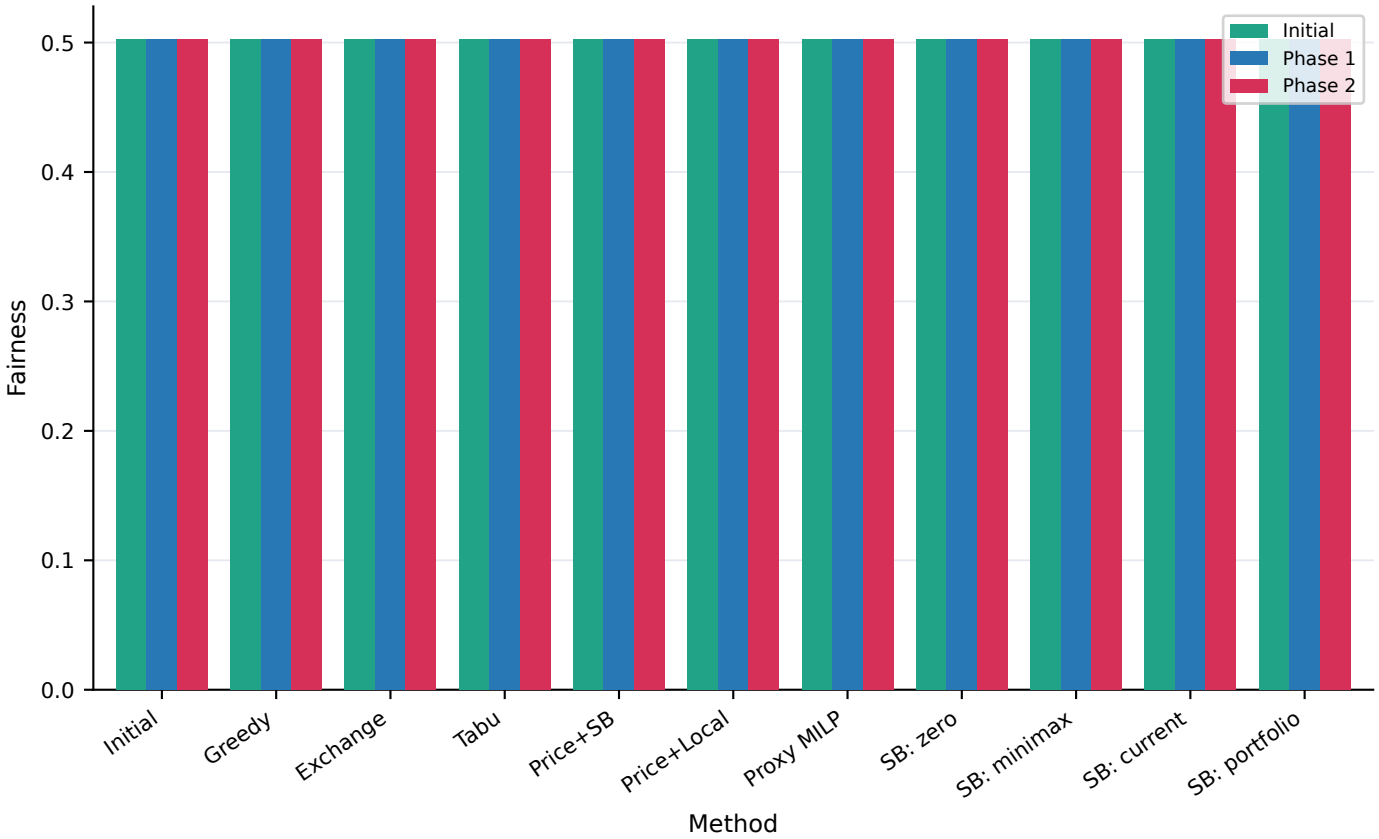
All SB work, including all portfolio members




small | first requested seed 20261105
Method execution status across seeds
Initial
Greedy
Exchange
Tabu
Price+SB
Price+Local
Price+MIQP
Proxy MILP
SB: zero
SB: minimax
SB: current
SB: portfolio
Method
20261105
20261106
20261107
Seed
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Physical availability (all candidates, before scheduling)
User
1
2
3
4
5
6
7
8
1
2
3
Block
1.0
0.8
0.6
0.4
0.2
0.0
0=unavailable, 1=budget exhausted before search, 2=failed, 3=completed (may stop at time limit)

small | first requested seed 20261105

Fairness at common initialization and each saved phase



Throughput at each phase (fairness is prioritized)

